

When Large Language Models Meet Law: Dual-Lens Taxonomy, Technical Advances, and Ethical Governance

PEIZHANG SHAO*, School of Law, China University of Political Science and Law, China and Zhejiang University of Finance and Economics Dongfang College, China
LINRUI XU*, School of Information Management for Law, China University of Political Science and Law, China and Department of Artificial Intelligence, Chung-Ang University, Republic of Korea
JINXI WANG, School of Law, China University of Political Science and Law, China
WEI ZHOU†, School of Information Management for Law, China University of Political Science and Law, China and School of Law, China University of Political Science and Law, China
XINGYU WU†, Department of Data Science and Artificial Intelligence, The Hong Kong Polytechnic University, China

This paper establishes the first comprehensive review of Large Language Models (LLMs) applied within the legal domain. It pioneers an innovative dual lens taxonomy that integrates legal reasoning frameworks and professional ontologies to systematically unify historical research and contemporary breakthroughs. Transformer-based LLMs, which exhibit emergent capabilities such as contextual reasoning and generative argumentation, surmount traditional limitations by dynamically capturing legal semantics and unifying evidence reasoning. Significant progress is documented in task generalization, reasoning formalization, workflow integration, and addressing core challenges in text processing, knowledge integration, and evaluation rigor via technical innovations like sparse attention mechanisms and mixture-of-experts architectures. However, widespread adoption of LLM introduces critical challenges: hallucination, explainability deficits, jurisdictional adaptation difficulties, and ethical asymmetry. This review proposes a novel taxonomy that maps legal roles to NLP subtasks and computationally implements the Toulmin argumentation framework, thus systematizing advances in reasoning, retrieval, prediction, and dispute resolution. It identifies key frontiers including low-resource systems, multimodal evidence integration, and dynamic rebuttal handling. Ultimately, this work provides both a technical roadmap for researchers and a conceptual framework for practitioners navigating the algorithmic future, laying a robust foundation for the next era of legal artificial intelligence.

CCS Concepts: • General and reference → Surveys and overviews; • Information systems → Data encoding and canonicalization; • Applied computing → Law.

*Both authors contributed equally to this research
†Corresponding Authors: Wei Zhou and Xingyu Wu

Authors' Contact Information: Peizhang Shao, School of Law, China University of Political Science and Law, Beijing, China and Zhejiang University of Finance and Economics Dongfang College, Haining, China; Linrui Xu, School of Information Management for Law, China University of Political Science and Law, Beijing, China and Department of Artificial Intelligence, Chung-Ang University, Seoul, Republic of Korea; Jinxi Wang, School of Law, China University of Political Science and Law, Beijing, China; Wei Zhou, cu0085859@cupl.edu.cn, School of Information Management for Law, China University of Political Science and Law, Beijing, China and School of Law, China University of Political Science and Law, Beijing, China; Xingyu Wu, xingyu.wu@polyu.edu.hk, Department of Data Science and Artificial Intelligence, The Hong Kong Polytechnic University, Hong Kong SAR, China.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
© 2025 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM 1557-7341/2025/6-ART111
<https://doi.org/XXXXXXX.XXXXXXX>

Additional Key Words and Phrases: Large language model, law, legal semantic, Toulmin argumentation framework, ethics

ACM Reference Format:

Peizhang Shao, Linrui Xu, Jinxi Wang, Wei Zhou, and Xingyu Wu. 2025. When Large Language Models Meet Law: Dual-Lens Taxonomy, Technical Advances, and Ethical Governance. *ACM Comput. Surv.* 37, 4, Article 111 (June 2025), 34 pages. <https://doi.org/XXXXXXX.XXXXXXX>

1 Introduction

The field of Artificial Intelligence and Law has evolved through three transformative decades since the inaugural issue of its flagship conference ICAIL in 1987 [7]. Early research focused on symbolic approaches including legal ontologies for knowledge sharing [72], rule-based reasoning systems, and argumentation frameworks. These established foundational paradigms but faced persistent challenges: ontological engineering struggled with semantic interoperability as conceptual mismatches hindered cross-system reuse [182], evidence reasoning remained fragmented between narrative and probabilistic methods without unified computational frameworks [17], and practical applications rarely moved beyond laboratory prototypes due to knowledge engineering bottlenecks [98]. This historical context illuminates why the evolution of legal artificial intelligence has transitioned from these symbolic expert systems [172] to contemporary data-driven neural models [10], marking a paradigm shift from logic-based formalisms to statistical learning approaches capable of handling law's inherent complexities. In summary, Early AI systems grappled with two fundamental constraints rooted in law's unique characteristics. First, the semantic richness of legal language demands context-aware interpretation beyond syntactic patterns—a challenge acutely felt in ontology engineering where attempts to formalize concepts like "reasonable doubt" revealed irreconcilable jurisdictional variations [145]. Second, the procedural rigidity of task-specific architectures proved ill-equipped for multi-step reasoning across evidentiary, normative, and factual dimensions [106, 142], as seen in isolated evidence models that combined arguments, stories, and Bayesian networks without theoretical integration [184]. While transitional technologies like domain-adapted embeddings (Law2Vec; [34]) and BiLSTM-based judgment predictors [35] offered incremental progress, their narrow task focus and limited generalization capacity fundamentally hindered end-to-end legal workflow integration. These limitations were particularly evident in real-world deployments where legal practitioners required systems capable of fluidly transitioning between statutory interpretation, precedent analysis, and evidentiary assessment—a holistic capability that remained elusive until the advent of large language models.

The emergence of transformer-based architectures has catalyzed a transformative shift in legal AI. Unlike predecessors, LLMs exhibit emergent capabilities—contextual reasoning, few-shot adaptation, and generative argumentation—that transcend traditional NLP pipelines [29]. These capabilities directly address long-standing gaps identified during AI & Law's first three decades: they overcome ontological fragmentation through contextual embeddings that dynamically capture legal semantics [100], unify evidence reasoning via coherent narrative generation that integrates probabilistic and argumentative elements [18], and enable practical deployment at scale by circumventing manual knowledge engineering bottlenecks [140]. This technological leap redefines the AI-Law nexus fundamentally. Where pre-LLM research optimized fragmented tasks—such as charge prediction via BiLSTMs [214] or statute retrieval using BM25 [150]—contemporary frameworks (GPT-4 [2], LLaMA-Law [41]) now enable unified architectures capable of handling end-to-end litigation and non-litigation workflows.

Crucially, the legal domain demands LLMs due to their unprecedented alignment with three core requirements that have challenged legal informatics since its inception. For complex text processing, traditional methods like rule-based summarization and SVM classification failed to capture

jurisprudential nuance—a limitation starkly evident in early lexical ontologies [145]. LLMs overcome this through abstractive summarization techniques like the ETA method [88] that preserve legal semantics, though they introduce hallucinations mitigated via knowledge-graph grounding architectures such as ChatLaw [42]. Regarding reasoning and argumentation, symbolic systems implementing Toulmin frameworks [175] lacked scalability—a gap documented in Walton’s argumentation schemes [187]. Modern LLMs enable generative warrant reasoning through specialized architectures like Lawformer [200] while leveraging retrieval-augmented generation (RAG) for evidentiary backing in COLIEE benchmarks [66], though qualifier calibration in sentencing prediction remains challenging [111]. Finally, for procedural augmentation, where pre-LLM tools operated in isolation (e.g., mediation outcome predictors [26]), contemporary systems facilitate human-AI collaboration in multi-agent negotiation scenarios [205], albeit raising ethical concerns about adversarial asymmetry originally identified in legal evidence studies [17].

This paper systematically summarizes three synergistic approaches essential for the effective deployment of LLMs in the demanding domain of legal practice. These approaches address fundamental challenges that have historically hindered legal AI systems and collectively enable transformative capabilities. First, context scalability—absolutely critical for processing the thousand-page case files commonplace in litigation—is achieved through advanced sparse attention mechanisms exemplified by architectures like Longformer [14]. This technical innovation directly overcomes the evidence integration bottlenecks identified in earlier hybrid Bayesian models [183], allowing LLMs to efficiently analyze vast evidentiary records that were previously computationally intractable. The ability to handle extensive context is foundational for tasks like comprehensive case summarization and identifying subtle precedents buried within lengthy documents, moving beyond the fragmented capabilities of earlier systems. Second, knowledge integration—crucial for grounding LLM outputs in authoritative legal principles and drastically reducing hallucination—is implemented via sophisticated hybrid architectures. Techniques such as mixture-of-experts (MoE) systems integrated with structured legal knowledge graphs, as demonstrated in models like ChatLaw [42], provide this essential grounding. This approach represents a significant evolution, extending the foundational ontology engineering principles established by pioneers like Visser and Bench-Capon [182] into the neural network era. By explicitly incorporating domain-specific knowledge structures, these systems enhance the reliability of outputs for critical tasks like legal reasoning and argument generation, ensuring fidelity to established legal doctrine where earlier AI often faltered. Third, evaluation rigor—an aspect long neglected in the deployment of practical legal AI systems [99]—is now being systematically addressed through the adoption of specialized, domain-relevant benchmarks. Frameworks like LawBench [56] and LexGLUE [36] establish standardized performance metrics tailored to legal tasks, moving beyond generic NLP evaluations. This rigorous assessment paradigm is vital for quantifying progress in complex legal applications such as judgment prediction, legal question answering, and precedent retrieval, providing the necessary evidence base for trust and adoption in professional settings. These three synergistic approaches—context scalability, knowledge integration, and evaluation rigor—collectively underpin three paradigm-shifting advancements that bridge persistent historical gaps in legal AI.

Despite these transformative advances, the LLM revolution introduces novel challenges that demand scholarly attention and technical innovation. Hallucination in legal claims manifests as spurious citations or normative fabrications, with cross-jurisdictional question answering systems exhibiting error rates as high as 58% [44]—a phenomenon echoing computational trust issues in expert testimony identified by Walton [186]. Mitigation strategies like retrieval augmentation [129] show promise but require further refinement. Explainability deficits create significant accountability gaps as black-box reasoning obscures decision pathways, perpetuating adoption barriers first noted in the 1990s [81] and partially addressed through syllogistic prompting techniques [47].

Jurisdictional adaptation remains problematic, with performance degradation in low-resource legal systems [117] necessitating specialized few-shot tuning approaches that extend ontological localization challenges documented by Leone et al. [100]. Perhaps most critically, ethical asymmetry emerges when disparities in LLM access exacerbate existing power imbalances between legal actors [193], requiring fairness-aware architectural interventions that respond to Walton's foundational concerns about argumentative justice [188]. These challenges collectively represent the "next frontier" in legal AI, demanding interdisciplinary solutions that integrate technical innovation with jurisprudential principles.

Previous reviews in the field of law and artificial intelligence have two main significant issues: On the one hand, there is the issue of comprehensiveness. Some works only focus on subtasks in the NLP domain [6, 8, 203], subtasks in the judicial prediction domain [43, 58], or subtasks in the IR domain [119, 128], which results in the review content often being insufficiently comprehensive. On the other hand, there is the issue of depth. They do not consider the legal problem itself, fail to take into account the ontological development process of legal artificial intelligence (or the ontological issues) [96], lack a technical review and large model application framework based on legal reasoning itself [165], or the articles themselves are relatively brief [149, 171]. Moreover, there is no review or discussion of the rules for legal professionals to use large models and their practical application areas.

This work presents the first comprehensive review of LLMs in legal contexts through a novel dual-lens taxonomy that integrates historical insights with contemporary breakthroughs. Our contributions advance the field in three significant dimensions:

- **Dual Lens Taxonomy Innovation:** Inspired by Calegari et al.'s work about legal reasoning [32], Section 3 establishes the first legal reasoning ontology framework aligning Toulmin's argumentation components (Data, Warrant, Backing, Claim) with LLM workflows, implementing Bex's evidence theory [19] at scale, while incorporating contemporary LLM breakthroughs with the evidence research of the past [181]; Section 4 mapping practitioner roles (lawyers, judges, litigants) to NLP tasks, extending user-centered ontology studies initiated by Francesconi and Araszkiewicz [59].
- **Scenario-Adaptive Deployment Framework:** Section 4 pioneers a role-centric framework for embedding LLMs across litigation (Section 4.1) and non-litigation (Section 4.2) workflows, addressing Lauritsen's call for "smarter tools" [98].
- **Ethical & Professional Mapping:** Section 5 presents the first systematic analysis of ethical challenges across practitioner roles (lawyers, judges, litigants), addressing both technological ethics (bias mitigation, hallucination controls) and legal professional responsibilities [59], extending user-centered ontology for LLM deployment [62].

As shown in Figure 1, the remainder of this paper is organized as follows. We first trace the technical evolution from small models to LLMs in Section 2, examining how neural architectures overcome knowledge representation barriers identified in early ontological studies. Section 3 analyzes LLM-enhanced legal reasoning through the Toulmin framework, extending legal LLMs tasks to the the ontology of legal reasoning. Next, Section 4 examines human-AI collaboration in litigation and alternative dispute resolution. Finally, Section 5 discusses ethical co-regulation frameworks and future directions in the era of LLM, incorporating lessons from three decades of applied research in legal technology. We have created a GitHub repository to index the relevant papers: https://github.com/Kilimajaro/LLMs_Meet_Law

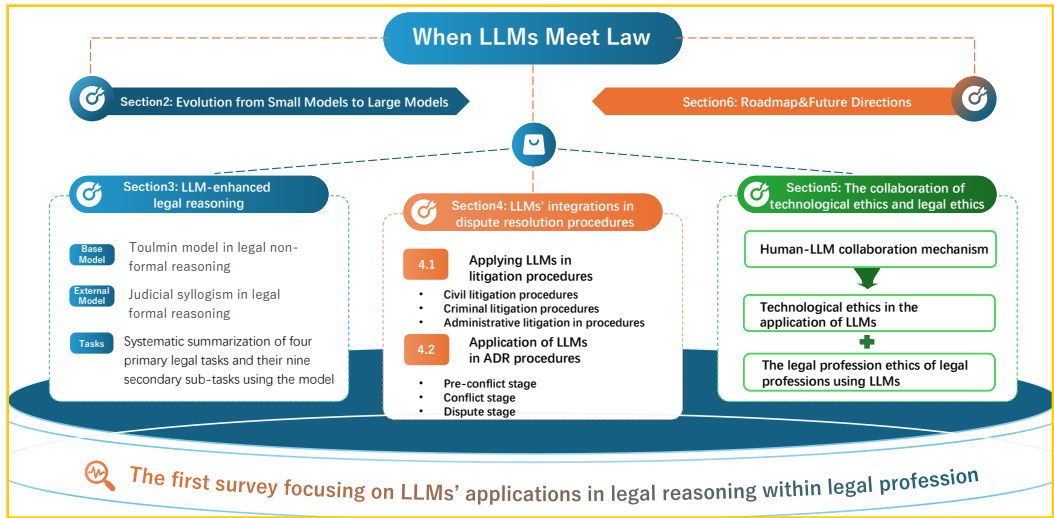


Fig. 1. A general framework for integrated research of LLMs and Law

2 Evolution from Small Models to Large Models

Section 2 traces the evolution of legal AI, contextualizing its progression from task-specific small models to versatile LLMs within broader NLP advancements. Section 2.1 outlines the technical trajectory of NLP, beginning with static embeddings (e.g., Word2Vec) and advancing through transformer architectures (e.g., BERT, GPT-3) that enable scalable, context-aware reasoning. Section 2.2 then examines how legal AI mirrored this shift, transitioning from early systems such as domain-adapted embeddings (Law2Vec) and BiLSTMs for clause segmentation to LLM-driven frameworks (Legal-BERT, LexGLUE) and generative models (e.g., GPT-3, ChatLaw) fine-tuned for legal tasks.

2.1 Evolution of NLP Models

The evolution of NLP models has been characterized by a paradigm shift from small-scale, task-specific architectures to large-scale, general-purpose frameworks [197]. Early advancements in distributed semantic representation laid the groundwork for modern language models. The introduction of word2vec by [123] marked a pivotal transition, enabling efficient learning of word embeddings through skip-gram and continuous bag-of-words (CBOW) architectures. These embeddings capture syntactic and semantic relationships, forming the basis for downstream NLP tasks. However, the limitations of static embeddings in handling polysemy and contextual nuances have spurred the development of dynamic, context-aware architectures. Subsequent innovations such as ERNIE [212] extended this paradigm by incorporating structured knowledge graphs through entity alignment and novel pretraining objectives, demonstrating significant improvements in knowledge-intensive tasks such as entity typing and relation classification through bidirectional fusion of textual and knowledge representations.

The architectural foundations of modern NLP were established by the transformer model [179], which introduced self-attention mechanisms to replace sequential processing, enabling parallelized training and superior long-range dependency modeling. Two distinct pretraining paradigms emerged: bidirectional encoding and autoregressive generation. The BERT model [49] pioneered

masked language modeling (MLM) through bidirectional transformer encoders, generating context-aware representations that revolutionized tasks such as question answering. Concurrently, the GPT series [151, 152] established the autoregressive paradigm, with GPT-1 using left-to-right language modeling on BooksCorpus and GPT-2 scaling to 1.5B parameters on WebText data, demonstrating zero-shot task transfer capabilities. This dichotomy was bridged by hybrid architectures such as BART [102], which combines bidirectional encoders with autoregressive decoders through denoising objectives (e.g., text infilling and sentence permutation), achieving state-of-the-art performance in both comprehension and generation tasks. Moreover, efficiency-driven innovations emerged: ELECTRA [38] replaced MLM with replaced token detection, achieving 4× faster convergence by training discriminators to distinguish real tokens from generator-produced substitutes, whereas ALBERT [97] reduced parameter counts through factorized embedding layers and cross-layer parameter sharing without compromising performance.

The scaling hypothesis reached its zenith with GPT-3 [29], whose 175B parameters and few-shot learning capabilities redefined model generalization, whereas the T5 framework [153] unified all NLP tasks as text-to-text transformations, pretraining on a 34B token C4 corpus with span corruption objectives. Notably, ERNIE 3.0 [170] integrated these directions through a hybrid architecture combining autoregressive flow and autoencoding paradigms, further enhanced by knowledge graph embeddings across 96 relation types, achieving 85.9% accuracy on Chinese open-domain QA—a 7.2% absolute gain over pure text-based models.

2.2 Legal Domain Adaptation Trajectory

The evolution of artificial intelligence in legal practice reflects broader technological trends in natural language processing, progressing from task-specific small models to versatile LLMs. Early efforts focused on addressing the unique challenges of legal texts—such as domain-specific terminology, syntactic complexity, and structured reasoning requirements—through tailored architectures and rule-based systems. These foundational approaches emphasized feature engineering and domain adaptation, as seen in the development of legal word embeddings and specialized recurrent neural networks. The emergence of transformer-based architectures and pretraining paradigms catalyzed a shift toward scalable, generalizable models capable of handling diverse legal tasks.

This subsection reviews landmark literature on the methodological evolution of core legal NLP applications, as illustrated in Fig. 2. By contrasting traditional techniques with contemporary LLM-driven approaches, it highlights pivotal studies that marked paradigm shifts in the field.

Foundations Stage (2017-2019) witnessed pioneering domain-specific architectures. Representative works include: Chalkidis et al.'s Law2Vec established legal word embeddings through neural networks, directly addressing lexical gaps in general-purpose models [34]. Nguyen et al. refined BiLSTM, showing hybrid approaches where legacy architectures could integrate BERT-derived features for complex tasks like legal discourse segmentation [134].

Transition Stage (2019-2021) marked the paradigm shift to pretraining. Chalkidis Legal-BERT became the cornerstone for transfer learning, repurposing transformer pre-training on case law corpora to capture jurisdictional semantics [35]. And Zhong et al. systematized how domain-adapted NLP pipelines improve legal workflows [214].

Expansion Stage (2021-2022) saw the transformer specialization scaling new complexities. Nguyen et al. developed a transformer-based approaches for the legal text processing task (LTP) in the COLIEE 2021 competition [131], while Chen et al. compared the random forests with deep learning in legal text classification tasks for the first time [37].

Breakthrough Stage (2022-2023) culminated in holistic LLM applications transcending task boundaries: Bommarito et al.'s Bar Exam GPT validated generative models' comprehensive legal reasoning [27]; while Xiao et al.'s long-document transformer (2022) directly addressed Chinese

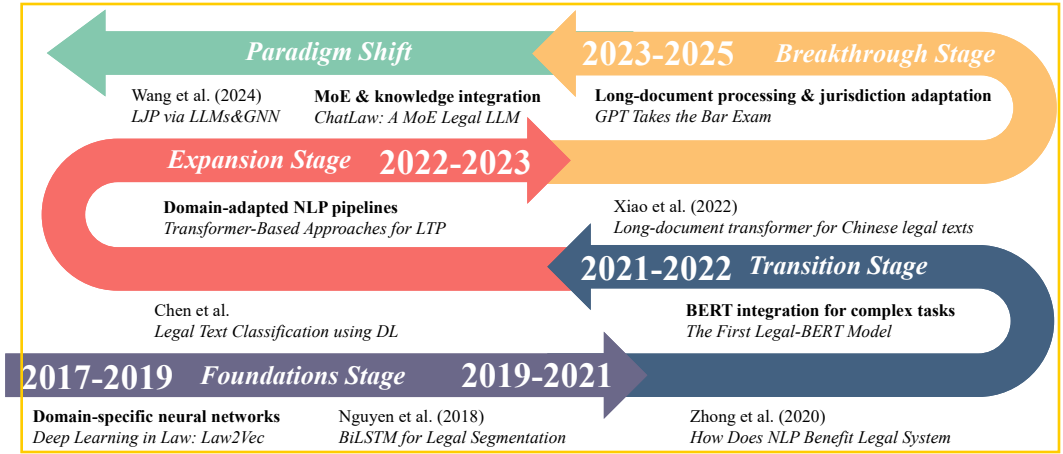


Fig. 2. Evolution of Legal NLP Models: From Task-Specific Small Models to Large Models Era

legal texts' structural peculiarities – proving that scaled attention mechanisms could overcome jurisdiction-specific document length challenges unmanageable by earlier RNN models [200].

Paradigm Shift(2023-2025) studies collectively signify the critical transition: Cui et al.'s ChatLaw (2023) MoE architecture exemplified domain-optimized scaling – where specialized legal expert modules dynamically route tasks within trillion-parameter systems [41, 42]; Wang et al.'s LLM-GNN fusion (2024) enabled multi-modal legal judgment prediction by combining textual reasoning with case relationship graphs [190].

The representative works outlined above trace the developmental trajectory of legal AI in the LLM era. A more comprehensive and systematic literature review, organized through the Toulmin model-based classification framework, will be presented in Section 3.

3 LLM-enhanced Legal Reasoning

The core of judicial work is the justification of judicial decisions [80]. The essence of this process lies in legal reasoning. In this case, Section 3 utilizes the legal reasoning model to elucidate the advancements in the legal LLMs field for systematicness, effectiveness and comprehensiveness. The application of this model highlights the latest theoretical interpretation of the formalization of legal reasoning within the context of the trend from symbolism to connectionism. Legal reasoning can be categorized into formal reasoning and substantive reasoning [101]. Compared with formal reasoning, substantive reasoning models are more suitable for refining various legal tasks in the legal process because of their effectiveness in confirming major and minor premises, achieving external justification. Therefore, Section 3 adopts the widely accepted Toulmin model in substantive reasoning to model the LLM-enhanced legal reasoning process and discusses in detail the corresponding legal tasks in each part of the Toulmin model and the research progress of their operation under the enhancement of LLMs.

3.1 LLM-assisted Legal Reasoning Structure

Section 3.1 introduces the method of decomposing large-scale model legal tasks on the basis of the Toulmin model, a substantive legal reasoning model. To complete a comprehensive legal reasoning process, both formal and informal logic play irreplaceable roles. Formal reasoning, which in the legal field is generally judicial syllogism, ensures the completion of internal justification [196].

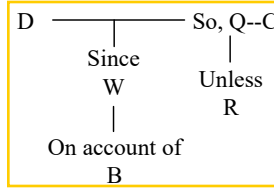


Fig. 3. Framework of Toulmin Model.

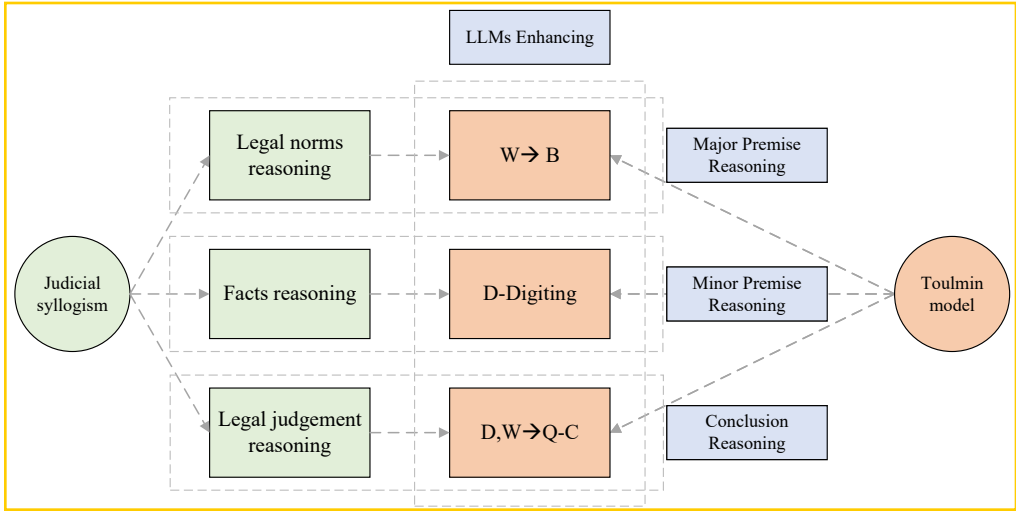


Fig. 4. Relationships between formal reasoning and informal reasoning.

Moreover, informal logic, which is commonly represented by the Toulmin model in the legal domain, ensures the completion of external justification. We first introduce the Toulmin model and elucidate the relationship between judicial syllogism and the Toulmin model.

As shown in Fig. 3, the Toulmin model dissects arguments into six key components—Data (D), Warrants (W), Backing (B), Qualifiers (Q), Rebuttals (R), and Claims (C)—and the entire argument process is a dynamic, adjustable structure that allows for adjustments and optimizations when faced with new evidence or arguments [175]. In the Toulmin model, the claim is the core of the argument, representing the main point that needs to be supported. To make the claim persuasive, it must rely on data, that is, specific facts, evidence, or examples that provide the foundation and support for the claim. The connection between the data and the claim is established through the warrant, which explains the logical relationship between the data and the claim, clarifying why the data can support the claim [93]. Moreover, the validity of the data often requires additional supporting material, which is the Backing. Backing provides further evidence or authoritative opinions for the warrant, enhancing the credibility of the argument. To avoid overly absolute statements, Qualifiers are introduced to specify the scope and strength of the claim, preventing overgeneralization or misinterpretation of the argument [79]. Finally, the rebuttal is used to address potential counterarguments or exceptions. It acknowledges the challenges that the argument may face and provides corresponding explanations or evidence to weaken these challenges.

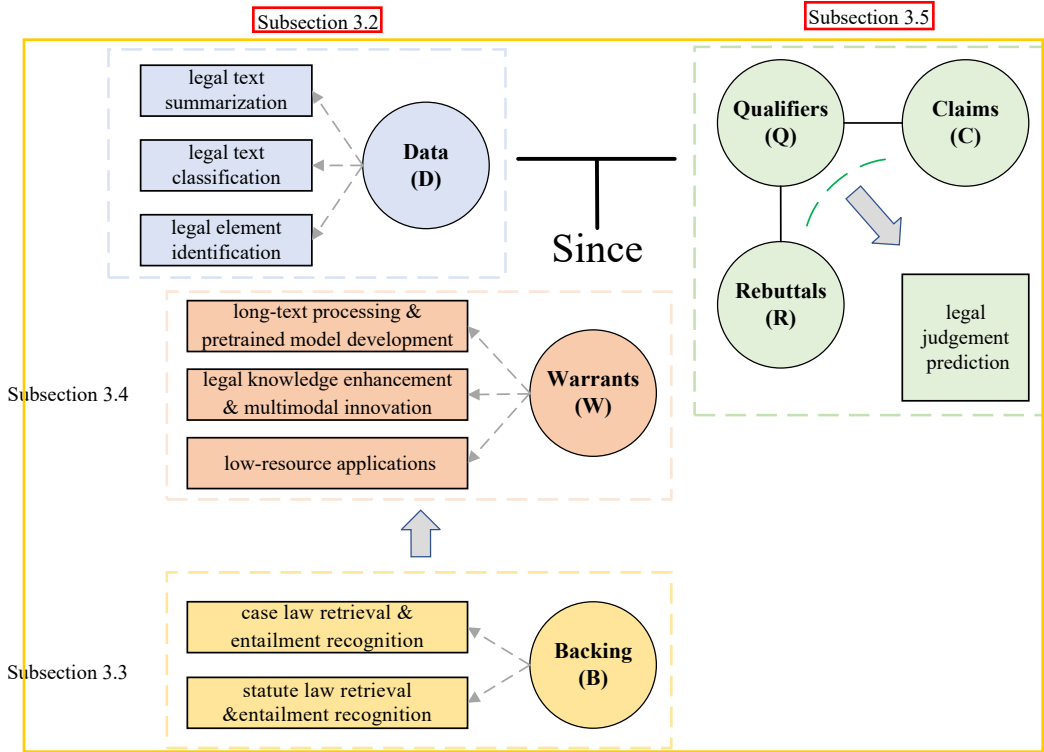


Fig. 5. Decomposition of Legal Reasoning Tasks Based on LLMs.

In judicial syllogism, legal rules are often the major premises that need to be discovered or determined. In such cases, the process of obtaining legal rules is a process of legal reasoning. At the same time, case facts are minor premises that involve a process of discovery or reasoning. Therefore, strictly speaking, legal reasoning encompasses three aspects: the reasoning of legal norms, the reasoning of facts, and the reasoning of judicial decisions [53], which can correspond to parts of the Toulmin model. As shown in Fig. 4, LLMs facilitate the three crucial aspects of judicial syllogism by assisting with various tasks in the informal reasoning of the Toulmin model, thereby achieving both external and internal justification and completing a comprehensive legal reasoning process. Therefore, this section focus on which tasks within each component of the Toulmin model can be accomplished by LLMs and the progress of relevant research to assess the extent to which LLMs can assist in legal reasoning.

As indicated in Fig. 5, for each of these components, LLMs are capable of performing relevant tasks to assist legal practitioners in conducting legal reasoning. The integration of LLMs into the legal domain has significantly advanced the capabilities of legal text processing, analysis, and prediction, offering robust ability for enhancing legal reasoning in the Toulmin framework. This section provides a detailed analysis of the roles and contributions of LLMs in each component, emphasizing their technical and strategic significance in the legal context.

Based on Fig. 5, we systematically introduce the following tasks. In Section 3.2, the focus is on data processing, where LLMs efficiently handle large-scale legal texts through summarization, classification, and element identification, exemplified by models that condense complex documents into actionable insights. In Section 3.3, legal backing is addressed through case law and statute

retrieval, supported by benchmarks to ensure that claims are grounded in authoritative precedents and statutory provisions. Section 3.4 emphasizes warrant optimization, leveraging long-text analysis, pretrained model fine-tuning, and multimodal integration to uncover implicit legal principles and enhance low-resource adaptability. Finally, Section 3.5 tackles argument formulation, enabling LLMs to generate and evaluate claims and qualifiers, and rebuttals while analyzing logical coherence to refine legal reasoning.

3.2 LLM-assisted Legal Data Processing

In this subsection, legal data processing tasks can be divided into three primary categories: legal text summarization, legal text classification, and legal element identification.

3.2.1 Legal text summarization task. Legal text summarization aims to extract key information from lengthy legal documents through automated technology, generating concise and accurate summaries to assist legal professionals in efficiently handling a large volume of legal texts. Legal text summarization research initially employed neural network methods such as word embeddings [49, 122, 144] and sentence embeddings [78, 91, 110] for semantic capture, laying foundations exemplified by work on Indian judgments [5]. However, these methods often struggled with context and generalizability. Reinforcement learning (RL) subsequently offered improvements, with advances including novel reward functions for deep modeling training [130]. The field has been significantly transformed by LLMs and related innovations: projects now leverage LLMs like GPT-4 via prompt engineering or fine-tuning methods such as LoRa, with memory-efficient training methods like gradient accumulation, gradient checkpointing [109], to directly generate summaries and extract information, especially when facing low-source situations [16, 48, 146], while novel architectures like the "extract-then-assign" (ETA) method address challenges of length and data scarcity through improved dataset construction [89]. To improve the evaluation, particularly for lengthy legal opinions, new dimensions such as argumentation coverage, event description accuracy, date accuracy, event capture completeness, and language quality have been introduced [54, 116].

3.2.2 Legal text classification task. The legal text classification task aims to classify legal documents efficiently and accurately through automated technology, thereby assisting legal professionals in quickly locating and processing relevant cases [26]. Although early legal text classification used primarily traditional machine learning, NLP models, and deep learning techniques - including random forests [37] and argument mining methods [142] - to process documents, the landscape changed significantly with the adoption of advanced transformer architectures and LLMs. Building on foundational transformer applications to legal issues [131], domain-specific pre-trained models like RoBERTa and BigBird were leveraged to classify cases from the European Court of Human Rights [87], and fine-tuned GPT-3 demonstrated effectiveness in legal rule classification [105]. LLM capabilities were further enhanced through task dependency, contrastive learning for case outcome prediction [177], and zero-shot transfer learning for generalizability to unseen legal articles [156]. Recent innovations include employing label models to integrate ChatGPT outputs and exploring prompt engineering for legal text entailment [21, 127], alongside hierarchical frameworks for classifying large unstructured legal documents [148], collectively demonstrating substantial progress in addressing legal text complexity.

3.2.3 Legal element identification task. The legal element identification task aims to extract key elements from legal documents, which is crucial for various legal applications, such as case analysis and legal information retrieval.

The development of techniques for this task has evolved from traditional machine learning models to more advanced deep learning architectures. Initially, Smuadu et al. explored the NER

Table 1. COLIEE Competition Task.

Category	Task	Description
Case Law Tasks	Task 1: Case Law Information Retrieval	Participants are required to retrieve relevant cases from a given case repository based on query cases. The challenge lies in identifying potential associations between cases rather than relying on simple pattern matching.
	Task 2: Case Law Entailment Recognition	Participants need to determine whether there is an entailment relationship between existing cases and unseen cases.
Statute Law Tasks	Task 3: Statute Law Information Retrieval	Participants are required to retrieve relevant legal provisions from statutes based on queries.
	Task 4: Statute Law Entailment Judgment	Participants need to determine whether a given legal statement is correct.

problem in Romanian and German legal domains via datasets such as RONEC and LegalNERo and employed models such as BERT, BiLSTM, and CRF [166]. This work laid the foundation for further research on legal element identification by demonstrating the effectiveness of domain-specific models. Yin et al. subsequently proposed a conversational mixture of expert large language models that was designed with specific prompts for case characteristics, utilized datasets such as CAIL2019 and a provincial theft case dataset, and employed tools such as ChatGLM3-6B and DoRA [82]. This study advances the field by introducing a more sophisticated and interactive approach to legal case element recognition. Furthermore, recent advancements have seen the development of approaches that directly design prompts and retrieval-augmented generation (RAG) systems on the basis of LLMs for legal element identification. These methods have demonstrated superior performance, as evidenced by their successful application in relevant tasks [154]. Such innovations underscore the growing potential of LLMs in enhancing the precision and efficiency of legal element extraction, marking a pivotal step forward in the integration of artificial intelligence within the legal domain.

Research in this area has clearly progressed from basic models to more complex and context-aware systems, with each study building upon the previous ones to improve the accuracy, efficiency of legal element identification and multi-domain legal text processing technologies [125].

3.3 LLM-assisted Legal Backing Digitizing

The backing digitizing task aims to match the applicable backings for case adjudications. In common law jurisdictions, backings are case precedents, whereas in civil law jurisdictions, backings are statutory provisions. This task is highly dependent on information retrieval and textual entailment technologies for legal texts. Since 2011, the COLIEE competition has focused on legal text information retrieval and recognition of legal provisions and case datasets [66]. Therefore, relevant research generally revolves around COLIEE competition tasks and datasets. The COLIEE competition usually consists of four tasks, which are divided into two major categories: case law and statute law, as shown in Table 1.

While traditional legal text retrieval and entailment relied on basic deep learning and information retrieval, struggling with semantic complexity and provision-case relationships, the advent of LLMs overcame these limitations through superior language understanding, leading current trends to focus on advancing LLM architectures/fine-tuning and strategically integrating them with traditional IR techniques for optimal domain-specific performance.

3.3.1 Early exploration and foundation: Combining foundational deep learning techniques with traditional information retrieval models (2018–2020). Initial development of legal information retrieval technology explored traditional models and basic transformers, exemplified by Tran et al.'s encoded summarization approach, which hierarchically encoded documents into a vector space for phrase scoring, validated on COLIEE 2018 data [176]. Building upon early exploration, researchers shifted focus to deep learning methods. Shao et al. demonstrated this potential in 2020 with the BERT-PLI model, leveraging BERT to capture semantic relationships at the paragraph level and aggregating these interactions to infer case relevance, achieving significant improvements on COLIEE 2019 [160]. Furthermore, hybrid approaches emerged, such as Rabelo et al.'s framework combining traditional similarity models for initial scoring with transformer models for entailment classification, highlighting the advantage of integrating both methodologies [150].

3.3.2 Development and expansion: Improving deep learning models and new frameworks (2021–2024). With the continuous development of LLMs and legal information retrieval technology, researchers have begun to explore more complex technologies and model architectures. Research in legal information retrieval has evolved from standalone deep learning models to increasingly sophisticated integrations with LLM. The initial approaches used custom-trained models such as BERT ensembles for statute/case law retrieval [158] and unsupervised frameworks such as CaseFormer for capturing case distinctions [169]. This foundation catalyzed a transition toward fusing deep learning with enhanced retrieval paradigms.

Subsequent innovations integrated transformers with traditional techniques like BM25 for legal entailment tasks [90], while attention mechanisms evolved into hierarchical architectures for long legal documents [132]. This period saw prolific hybrid systems combining ordinal keyword IR with BERT-based retrieval [204], and merging BERT with graph networks and custom encoders [191]. Gleichzeitig, a critical evaluation emerged of the performance of the method [85] and the efficacy of pre-training efficacy of pre-training [213], establishing benchmarks for complex model architectures.

3.3.3 Innovation and future outlook. Recent innovations in legal information retrieval span three key dimensions. In retrieval frameworks, researchers propose multi-stage approaches [128] integrating LLMs with traditional methods for statute and case law retrieval [129], alongside methods improving accuracy through fine-grained fact-provision matching [63] and optimized retrieval-relevance ranking pipelines [103]. Addressing cross-domain optimization and user diversity, studies focus on reducing misinformation via knowledge-enhanced verification [124], mitigating positional bias in RAG systems [40], calibrating confidence for ambiguous queries [162], and leveraging user attention for relevance judgment [161]. Model structure innovations introduce GNN-based representations (CaseGNN) for capturing legal structural semantics [173] and causal relationship learning frameworks for provision selection [190]. Concurrently, data augmentation and model fusion techniques, employing contrastive learning, zero-shot prompting [31], model ensembles [133, 185], entity-aware tuning [139], and interpretable Chain-of-Thought prompting [60], significantly enhance performance.

Future research must prioritize enhancing robustness across diverse legal domains and user backgrounds [124, 161], addressing challenges like cross-jurisdictional generalization and bias mitigation [40]. Advancing model interpretability through techniques like CoT [60] and structurally-aware representations [173, 190], coupled with deeper integration of domain knowledge for semantic alignment [63], will be crucial. Further exploration into sophisticated LLM fusion strategies [31, 133, 185] and reliable verification paradigms [124, 162] holds significant potential to improve the accuracy, reliability, and practical utility of legal information retrieval systems.

3.4 LLM-assisted Legal Warrant Reasoning Generation

Legal warrant reasoning generation is crucial for extracting and reasoning with backings that support credible legal arguments. As legal texts grow in complexity, the ability to generate warrants based on known backings becomes a key challenge. This subsection outlines the development of LLM-assisted warrant reasoning in three aspects: long-text processing and legal domain pretraining, legal knowledge and multimodal integration, and applications in low-resource settings with future outlooks.

3.4.1 Long-text processing and pretrained model development in the legal domain. Long document processing is a foundational challenge in legal NLP. Lawformer [200], based on Longformer, demonstrated effective handling of long Chinese legal texts and outperformed prior models in tasks like judgment prediction. Its release as part of LegalPLMs provided a strong pretraining baseline, though it lacked explicit legal knowledge integration. Later models focused on domain adaptation. Lawyer LLaMA [83] fine-tuned LLaMA for legal Q&A, reducing hallucinations but limited by narrow training data. DISC-LawLLM [205] introduced syllogistic prompting and retrieval-augmented mechanisms, signaling a shift to task-oriented legal LLMs. Standardized benchmarks emerged to assess these models. LawBench [56] evaluated Chinese legal tasks using OpenCompass, while LEGALBENCH [73] covered 162 cross-jurisdictional reasoning tasks. These benchmarks revealed limitations of general LLMs and set evaluation standards for legal AI [56, 73].

3.4.2 Legal knowledge enhancement and multimodal method innovation. In 2024, structured legal knowledge integration became key. ChatLaw [42] employed a MoE framework and knowledge graphs, reducing hallucinations by 38% and improving citation-based reasoning. CBR-RAG [195] combined precedent retrieval with generation, boosting cross-jurisdictional Q&A accuracy by 17.3%. InternLM-Law [57] achieved state-of-the-art LawBench performance through two-stage fine-tuning. Ghosh et al. [65] combined LLaMA-2 with expert-curated graphs, achieving 91% precision in Indian legal document tasks, underscoring the value of human-in-the-loop systems. Hallucination analysis matured with Large Legal Fictions [44], identifying 42% error rates across 12 jurisdictions, and enabling mitigation evaluation via the Harvard Dataverse. Gray et al. [70] enhanced empirical legal research using GPT-4 for factor annotation, tripling efficiency with 98% consistency.

3.4.3 Low-resource scenario applications and future technical directions. LLMs have shown promise in low-resource contexts. Maree et al. [117] used ChatGPT and LlamaIndex for legal Q&A in Palestinian cooperatives, achieving 83% answer relevance. Deng et al. [47] introduced the SLJA Dataset, validating ChatGLM's effectiveness on 11,239 annotated criminal cases. Future directions include: (1) multimodal integration of text, knowledge graphs, and evidence [195, 195] and improvement of explainability, and resistance to AI-generated content interference using new architecture [208]; (2) legal-specific generative pretraining to address data sparsity [42, 83, 200, 205, 215]; and (3) evolving evaluation systems such as LegalBench and 's multilingual expansion [73]. These trends signal a shift from supportive tools to robust legal decision-support systems.

3.5 LLM-assisted Legal Judgment Prediction with Qualifiers

In this subsection, we present an overview of the evolution and current landscape of legal judgment prediction (LJP). LJP has evolved from early statistical methods such as SVMs and logistic regression [168] to deep learning models utilizing multistage case representation [112, 113, 147], further enhanced by reinforcement learning [111], multitask learning [84, 201, 202], and knowledge-augmented techniques via event extraction [58], contrastive learning [107, 210], and logic-rule injection [61]. While these models improved prediction accuracy and interpretability, they often struggled with generalization across languages and legal systems. The introduction of LLMs

marked a turning point in LJP research, enabling models to better understand long-form legal texts and reason over complex legal structures, as evidenced by benchmarks like LexGLUE [36] and multilingual datasets such as those from the Swiss Federal Supreme Court [136] and LBOX OPEN [86], which laid the groundwork for evaluating pretrained models in legal contexts.

In the initial investigations of LLM-aided LJP, researchers concentrated on benchmarking fundamental performance and intrinsic limitations in statutory prediction and outcome forecasting, exposing both potential and challenges in real-world judicial settings. Shaurya Vats et al. [180] applied LLMs to Indian Supreme Court cases, observing superior statutory prediction accuracy but inferior outcome prediction compared to baselines, while uncovering gender and religious biases in model outputs. Ruihao Shui et al. [163] proposed a practical LLM-based baseline solution, systematically evaluating LLM capabilities and demonstrating that analogous cases and label candidates boost performance, though weaker LLMs failed to exploit advanced information-retrieval systems; collectively, these studies highlighted performance bottlenecks and social-bias risks, furnishing an empirical foundation for subsequent refinements.

To address these bottlenecks, scholars devised novel frameworks for enhanced accuracy and interpretability through structured reasoning and model fusion. Shubham Kumar Nigam et al. [135] employed LLMs like GPT-3.5 Turbo in Indian judicial contexts, simulating realistic courtroom settings with case facts and precedents to reveal superior performance and introduce clarity and relevance metrics. Chenlong Deng et al. [46] proposed the ADAPT framework, decomposing facts and distinguishing charges to improve accuracy on ambiguous cases. Xia Yangbin et al. [199] introduced the LLG-Judger framework combining LLMs with graph neural networks. Zhang Yue et al. [211] developed a framework enhanced by first-order logic rules and comparative learning, and Bin Wei et al. [192] proposed a neuro-symbolic framework integrating logical rules with deep learning for civil-case adjudication—collectively marking the evolution toward refined reasoning and robustness in intricate scenarios.

In advanced analysis, researchers optimized practical deployment issues like information retrieval and noise suppression to improve stability and consistency. Ruihao Shui et al. [163] re-evaluated LLM performance, confirming that analogous cases and label candidates enhance accuracy but noting retrieval systems alone may outperform combined approaches; follow-up experiments showed label-candidate augmentation improves consistency, while noisy cases disproportionately degrade smaller models, underscoring the need to balance retrieval efficiency with noise mitigation. Recent efforts have directly integrated LLMs into LJP pipelines, such as Wang et al.'s LegalReasoner framework [189], a multistage architecture leveraging structured legal knowledge to enhance accuracy, interpretability, and adaptability to jurisdictional variations. Beyond binary prediction, LLMs now aim for qualified judgment outcomes using frameworks like the Toulmin Model to generate contextually grounded conclusions, such as nuanced sentencing decisions.

Looking forward, LLM-assisted LJP will benefit from cross-disciplinary collaboration integrating AI, law, and social sciences, as highlighted in surveys by Cui et al. [43] and Medvedeva et al. [120], emphasizing needs for richer datasets, multilingual resources, and improved interpretability to evolve LLMs into robust reasoning agents. In 2025, novel methods have been approached [159], such as JuriSim [75] and ILJR [104], incorporating the knowledge of judicial trial logic and other multi-source knowledge to manage the LJP task.

3.6 Toolbox and Datasets

We summarize the legal LLM tools and datasets mentioned in Section 3 in Table 2 and Table 3:

Table 2. Legal LLM Toolbox

Name	Type	Creator	Reference
Law2Vec	Word Embedding	Chalkidis et al.	[34]
Legal-BERT	Pretrained Model	Chalkidis et al.	[35]
MVE-FLK	Multi-task Legal LJP	Yang et al.	[202]
Lawformer	Long-text Processor	Xiao et al.	[200]
ChatLaw	MoE Architecture	Cui et al.	[42]
LawGPT	Fine-Tuning Model	Zhou et al.	[215]
DISC-LawLLM	Reasoning Framework	Yue et al.	[205]
InternLM-Law	Domain-specific LLM	Fei et al.	[57]
Lawyer LLaMA	Legal QA System	Huang et al.	[83]
CaseGNN	Graph-based Retrieval	Tang et al.	[173]
LegalReasoner	Judgment Predictor	Wang et al.	[189]
ADAPT	Fact-charge Decoupler	Deng et al.	[46]
LLG-Judger	GNN-LLM Fusion	Yangbin et al.	[203]
CBR-RAG	Precedent Retriever	Wiratunga et al.	[195]
ETA Method	Abstractive Summarization	Jain et al.	[89]
BERT-PLI	Statute Retriever	Shao et al.	[158]
Caseformer	Unsupervised Pretraining	Su et al.	[169]
MLMN	Provision Matcher	Ge et al.	[63]
ContractMind	Agreement Drafting	Zeng et al.	[209]
Syllogistic Prompting	Reasoning Enhancer	Deng et al.	[47]
ERNIE 3.0	Knowledge-enhanced LLM	Sun et al.	[170]
Longformer	Sparse Attention	Beltagy et al.	[14]
ELECTRA	Efficient Pretraining	Clark et al.	[38]
CTM	Judgement Prediction Framework	Liu et al.	[107]
BM25+Longformer	Joint Learning Approach	Nguyen et al.	[133]
Multi-task Learning	Legal Named Entity Recognition	Smadu et al.	[166]
Causality-inspired Method	Relational Reasoning	Wang et al.	[190]
JuriSim	Dual Residual Attention for LJP	He et al.	[75]
ILJR	Interpretable Legal Judgment Reasoning	Liu et al.	[104]
RoSILC-RS	Robust Similar Legal Case Recommender System	Zeng et al.	[208]

Table 3. Legal LLM Datasets

Name	Task	Creator	Reference
COLIEE	Legal Information Retrieval	Goebel et al.	[66]
LawBench	Summarization, Judgment Prediction, QA	Fei et al.	[56]
LEGALBENCH	Legal Reasoning Tasks	Guha et al.	[73]
LexGLUE	Multi-task Legal NLP	Chalkidis et al.	[36]
CUAD	Contract Review, Obligation Identification	Hendrycks et al.	[77]
CAIL	Legal Element Recognition, Judgment Prediction	Hua et al.	[82, 190]
Swiss-Judgment-Prediction	Multilingual Judgment Prediction	Niklaus et al.	[136]
LBOX OPEN	Korean Legal Language Understanding	Hwang et al.	[86]
SLJA Dataset	Syllogistic Reasoning, Legal Analysis	Deng et al.	[47]
RealToxicityPrompts	Toxicity Detection, Bias Evaluation	Gehman et al.	[46]
RONEC and LegalNERo	Legal Named Entity Recognition	Smadu et al.	[166]
UK Case Law Dataset	Multi-task Legal NLP	Izzidien et al.	[39]
LAiW	IR, Legal Reasoning, Legal Application	Dai et al.	[45]
RONEC	Romanian Named Entity Corpus	Dumitrescu et al.	[52]
LegalDocML	Naming Convention	Dimyadi et al.	[51]
LEXam	Legal Reasoning	Fan et al.	[55]
NLJP	LJP, Classification	Chalkidis et al.	[35]

4 LLMs' Integrations in Dispute Resolution Procedures

Section 4 explores how LLMs enhance both litigation and nonlitigation legal procedures, which resolve disputes through judicial processes in courts (governed by strict norms) or flexible alternatives like negotiation and arbitration, respectively. These models assist professionals in approximating legal "standard answers," with data-driven capabilities that inspire and assist, surpassing traditional tools [118]. This section discusses integrating LLMs into these workflows, positioning human professionals as critical intermediaries for connecting AI to legal practice through human-computer interaction.

4.1 Applying LLMs in Litigation Procedures

4.1.1 Applying LLMs in civil litigation procedures. In civil litigation procedures, LLMs provide comprehensive and efficient assistance to judges, lawyers, parties, and other litigation participants through the various components of the Toulmin model, significantly enhancing the quality and efficiency of legal argumentation.

For judges, whose responsibilities include presiding over trials, ensuring fairness, and making fact-based rulings, LLMs can rapidly distill case facts and key legal provisions through legal text summarization and legal element identification techniques at the D part of the Toulmin model [175]. This helps judges efficiently grasp the overall picture of the case. For example, LLMs can summarize trial records, evidence materials, and legal provisions, providing judges with a clear factual framework and legal basis. In the W part of the Toulmin model, LLMs can enhance legal knowledge to assist judges in analyzing the logical relationship between case facts and legal provisions, ensuring accurate application of the law in their rulings [114]. Additionally, in terms of B, LLMs can quickly retrieve relevant precedents and legal provisions to provide authoritative support for judges' rulings, thereby enhancing the persuasiveness of their decisions [66, 74].

Lawyers provide legal advice, draft documents, and present arguments in court. LLMs enhance their argumentation by summarizing legal texts, identifying key elements, and extracting relevant provisions, streamlining case preparation and document drafting [165]. In the argumentation stage, LLMs, with their legal knowledge enhancement and multimodal innovation technologies, help lawyers construct logically rigorous legal arguments, ensuring that their claims are closely linked to legal bases [42]. Moreover, LLMs can provide rich precedential and legal support through case law and statutory retrieval techniques, enhancing the credibility of their arguments [129].

Parties in civil litigation have the right to present their cases and evidence. LLMs can help parties better understand and express their claims by summarizing and refining their statements, enabling clear articulation of case facts and legal demands. Additionally, LLMs can provide basic legal knowledge through legal education functions, enabling parties to better protect their rights during the litigation process [65].

In civil litigation, LLMs enhance efficiency and precision. They help judges understand case facts, identify legal provisions, and apply precedents by analyzing legal texts. LLMs ensure judgments are logically coherent and grounded in law. For lawyers, LLMs streamline case preparation, from drafting pleadings to constructing arguments. Lay litigants benefit from LLM-generated summaries to articulate claims clearly. LLMs also assist witnesses and experts in preparing coherent testimonies. By bridging the knowledge gap, LLMs promote transparency and fairness in civil disputes.

4.1.2 Applying LLMs in criminal litigation procedures. The criminal litigation process involves various participating roles, each fulfilling specific functions and responsibilities. In the criminal justice systems of countries worldwide, key participants typically include judges, prosecutors, juries, defendants, defense attorneys, and witnesses such as expert witnesses. Judges preside over trials and apply the law based on the facts of the case to ensure the fairness and legality of the proceedings.

Prosecutors initiate public prosecutions for criminal cases on behalf of the state, present charges, and provide sufficient evidence to substantiate the alleged crimes. Defense attorneys represent defendants, offer legal counsel, draft legal documents, and advocate for their clients. Additionally, witnesses provide testimony or professional opinions to assist fact-finders in determining the truth of the matter at issue [71].

In criminal litigation, LLMs provide targeted support to prosecutors and defense attorneys by enhancing legal argumentation through the Toulmin framework. Prosecutors leverage LLMs to draft indictments by summarizing facts and identifying essential crime elements [165], construct robust legal arguments connecting claims to statutory bases [42], and bolster arguments through precedent retrieval for evidential support [129]. Defense attorneys utilize LLMs to rapidly identify case disputes and core defense strategies [70], critically analyze prosecution evidence to expose weaknesses in collection or authenticity [195], discover exculpatory evidence through database mining [117], and generate targeted defense arguments grounded in facts and statutes [47]. Additionally, LLMs assist witnesses in organizing testimony [173] and experts in rapidly accessing domain knowledge [60].

By applying the Toulmin model across these functions—structuring claims (C), grounding (D), warrants (W), backing (B), and rebuttals (R)—LLMs enhance argumentative rigor, factual relevance, and legal sufficiency. This deep integration ultimately improves procedural fairness and advances the pursuit of legally sound outcomes in criminal justice [53], while equipping all participants with more effective tools for truth-seeking.

4.1.3 Applying LLMs in administrative litigation procedures. In administrative litigation—distinguished by its focus on overseeing government power and placing the burden of proof on agencies—LLMs provide specialized support aligned with its unique objectives. For judges, they streamline review by summarizing administrative decisions and enforcement records while pinpointing critical legal elements like procedural compliance and evidentiary sufficiency [54], and enhance rulings through precedent retrieval that analyzes judicial reasoning in analogous regulatory contexts [148].

LLMs uniquely empower litigants: Plaintiffs gain clarity by distilling agency decisions and predicting outcomes through case law analysis [129], while government defendants leverage LLMs to meet strict proof burdens by rapidly generating compliance arguments (e.g., statutory adherence) and countering procedural challenges [47]. Lawyers utilize LLMs to detect procedural flaws, clarify fact-law connections for conditional statutes (e.g., licensing rules), and strengthen arguments against administrative overreach [154, 165].

Crucially, LLMs generate conclusive, jurisdictionally tailored rulings (e.g., "Action unlawful due to Article X violation") by synthesizing factual, legal, and compliance criteria [56]. This capability uniquely advances administrative litigation's core mission: balancing state authority with individual rights through rigorous legal-technical scrutiny absent in civil/criminal domains.

4.2 Application of LLMs in Alternative Dispute Resolution Procedures

In this section, we explore the integration of LLMs into Alternative Dispute Resolution (ADR) procedures across the three-stage nonlitigation framework (pre-conflict, conflict, dispute) [126]. At the pre-conflict stage, LLMs enhance preventive services like contract review and legal advice by automating clause analysis, identifying risks, and providing proactive insights to prevent disputes. During the conflict stage, LLMs optimize negotiations through real-time risk assessments, automated drafting, and e-discovery for electronic evidence management. In the dispute resolution stage involving third parties (mediation or arbitration), LLMs assist mediators by generating structured agreements and support arbitrators through accelerated legal research, precedent analysis, enforceable award drafting, and ensuring cross-jurisdictional compliance. Collectively, LLMs streamline ADR workflows, improve decision-making, and promote efficient resolutions [137].

4.2.1 Applying LLMs in the pre-conflict stage. In legal practice, the pre-conflict stage is crucial. During this phase, preventive measures and proactive legal advice can reduce future disputes. Legal services at this stage include contract review, due diligence, and prelitigation consultation. LLMs enhance efficiency by automating clause analysis, identifying ambiguous clauses, and providing real-time feedback. In due diligence, LLMs extract critical information, automate compliance assessments, and forecast financial implications. In prelitigation consulting, LLMs offer comprehensive legal guidance to mitigate risks.

Contract review is essential in legal practice, aiming to ensure clear, enforceable clauses aligned with best practices and minimizing disputes. Traditional methods rely on manual lawyer analysis, which is time-consuming, labor-intensive, and prone to errors. LLMs have transformed contract review by improving efficiency, enhancing clause analysis accuracy, increasing trust in AI recommendations, and offering real-time feedback for revisions. LLMs can quickly analyze contracts, identify ambiguous clauses, compliance issues, and cross-reference clauses with legal databases and standards [209]. For example, The CUAD dataset provides high-quality labeled data for applying LLMs in contract review, enabling models to better understand and process contract texts [77]. Through dynamic interactive design, like the ContractMind system, LLMs can provide real-time AI recommendations, legal clauses, and case support, helping legal professionals efficiently review contracts by calibrating their trust in AI [209]. LLMs can more accurately identify obligations, permissions, and prohibitions in contracts using CNNs and LSTM, providing legal professionals with comprehensive contract analysis [69]. These technological advancements have improved contract efficiency and accuracy, enhanced review quality and reliability through automation, and provided legal professionals with a more efficient solution [121].

During the due diligence process, ensuring transaction security and legality is the core objective. The application of LLMs is reshaping this traditional procedure by improving efficiency and accuracy. LLMs use methods like conditional random fields (CRFs) to process large volumes of unstructured data, such as financial reports and legal documents, enhancing information extraction [155]. AI-driven frameworks employ machine learning algorithms, such as random forests, for financial data analysis to predict future company performance and assess financial health [13]. NLP automates legal compliance checks by reviewing documents and identifying risks, improving review efficiency and accuracy [13]. Additionally, LLMs analyze complex legal issues raised by administrative agencies and provide response suggestions, which are crucial for identifying legal risks [47]. These technologies enhance due diligence efficiency and accuracy while providing decision-makers with deeper insights, ensuring secure and legal transactions.

In the pre-conflict stage, legal advisory services provide proactive guidance to prevent disputes. Due diligence involves evaluating a client's legal status, risks, and compliance with laws. This includes reviewing contracts and analyzing legal documents. Traditional consulting is often limited by lawyer availability and expertise, leading to inconsistent advice. LLMs offer consistent, high-quality legal insights by accessing extensive legal knowledge and precedents [41]. LLMs analyze complex scenarios, interpret regulations, and generate tailored legal recommendations [205], enhancing advisors' ability to deliver comprehensive advice [56]. Moreover, LLMs significantly enhance real-time legal research and analysis capabilities, enabling advisors to promptly address client inquiries and emerging legal challenges [57]. By integrating LLMs into legal advisory services, law firms can proactively identify and mitigate potential legal risks, minimizing disputes and fostering a more knowledgeable and resilient client base [83].

4.2.2 Applying LLMs in the conflict stage. In the conflict stage, the dispute can still be resolved through negotiation, a self-help remedy that does not involve third parties and is the least costly method. Negotiation often involves collecting and analyzing electronic evidence, so e-discovery is

discussed in this subsection. As legal matters grow more complex and technology advances, LLMs are becoming key tools in the legal field. In e-discovery, LLMs enhance data processing efficiency and accuracy through technical improvements and hybrid methods while addressing ethical and privacy concerns. During negotiations, LLMs draft legally compliant agreements, interpret clauses, and assess legal risks in real-time, helping negotiators understand terms and create mutually beneficial agreements.

In the field of e-discovery, LLMs have emerged as a key solution to address data volume challenges and improve efficiency. Research has focused on three areas: technical optimization and ethical/privacy protection. For technical optimization, instruction tuning and techniques like LoRA/Q-LoRA enhance open-source model performance, surpassing commercial models in document relevance tasks and providing strong legal support for e-discovery [141]. Ethical concerns are addressed through diverse training data and bias audits, while blockchain integration enhances data integrity and evidence chain management [167] to support clients' needs for non-discrimination and data accuracy in e-discovery processes. These advancements show that LLMs improve e-discovery efficiency, accuracy, and ethical handling of legal data.

In negotiations, negotiators must draft clear settlement agreements. LLMs can help create legally compliant drafts with precise language and accurate content [83]. Additionally, LLMs provide insights into relevant laws, assisting negotiators in assessing the legal risks and benefits of proposed terms [56].

Negotiators must draft clear, legally sound agreements that reflect settlement terms. Well-drafted agreements reduce ambiguity, prevent disputes, and ensure enforceability. To do this effectively, negotiators need a strong grasp of legal terminology, contract structures, and relevant laws. LLMs can help by generating contract drafts aligned with legal standards and best practices. By analyzing extensive legal databases, LLMs ensure precise, unambiguous language tailored to the negotiation context. For instance, in agreements involving intellectual property, LLMs include appropriate clauses on ownership, licensing, and infringement protection [42]. Additionally, LLMs identify inconsistencies and suggest improvements to enhance enforceability and minimize postnegotiation disputes.

Beyond the drafting process, negotiators must possess a comprehensive understanding of the legal implications associated with the terms under negotiation to make well-informed decisions and anticipate potential risks. Legal agreements often include complex clauses that can lead to unintended consequences if not thoroughly evaluated. In this context, LLMs function as an indispensable tool by delivering real-time legal insights and risk assessments. For example, when evaluating a limitation of liability clause, an LLM can rapidly analyze relevant legal precedents and statutory provisions to identify common pitfalls and enforcement challenges [48]. Additionally, LLMs benchmark proposed terms against industry standards, regulatory requirements, and judicial precedents, assisting negotiators in assessing the fairness and feasibility of the terms. By providing data-driven legal risk analysis, LLMs enable negotiators to construct agreements that are mutually advantageous and legally resilient, ensuring all parties clearly understand their respective rights and obligations before finalizing the deal.

4.2.3 Applying LLMs in the dispute stage. The application of LLMs in dispute resolution can enhance traditional legal processes by improving efficiency, accuracy, and fairness. As disputes grow more complex and diverse, LLMs provide advantages in arbitration and mediation by automating tasks, offering legal insights, and ensuring outcomes align with standards and interests.

(1) **Applying LLMs in arbitration:** In arbitration, LLMs significantly support arbitrators by drafting enforceable awards with precise, legally compliant language to enhance enforceability [61]. They expedite legal research by rapidly retrieving and analyzing relevant laws, regulations, and

precedents—crucial for complex cases and aligned with trends showing 73% of legal professionals planning AI adoption for such tasks [92, 200]. Furthermore, LLMs analyze case details against legal benchmarks, identify jurisdictional risks in international disputes, and provide actionable mitigation insights [165], thereby improving decision equity, process efficiency, and overall fairness in arbitration outcomes.

(2) Applying LLMs in mediation: Mediation is a process that heavily depends on effective communication, sound legal reasoning, and the creation of structured settlement agreements. LLMs can substantially enhance the efficiency and fairness of mediation by supporting mediators in drafting clear, legally robust mediation agreements. These agreements must precisely reflect the interests and commitments of both parties while ensuring enforceability under applicable legal frameworks. LLMs are capable of generating well-structured draft agreements that adhere to established legal principles, minimize ambiguity, and mitigate the risk of future disputes. Additionally, they can analyze previous mediation cases with similar contexts to recommend widely accepted clauses, thereby assisting mediators in crafting balanced and comprehensive agreements that satisfy all stakeholders involved [68].

Another pivotal aspect of mediation is facilitating effective and constructive communication between the disputing parties. Mediators ensure both sides understand their legal positions, risks, and options. LLMs assist by summarizing pertinent laws, analyzing case law, and explaining complex legal terms impartially. This enables mediators to provide precise, objective information, promoting informed decision-making. Additionally, LLMs help structure sessions by creating agendas, encapsulating key points, and proposing compromise solutions based on data and patterns. By offering real-time insights and structured support, LLMs enhance mediators' capacity to guide parties toward resolutions [65].

LLMs can enhance post-mediation processes by automating the review of settlement agreements. They ensure alignment with laws and stakeholder interests by cross-checking terms against regulations, identifying risks, and proposing refinements. In cross-border cases, LLMs conduct comparative legal analyses for compliance across jurisdictions. By integrating LLMs, mediators improve efficiency, fairness, and the durability of settlements. Additionally, LLMs provide real-time legal insights and plain-language summaries to facilitate informed discussions and build trust in the mediation process [76].

5 The Collaboration of Technological Ethics and Legal Ethics

This section analyzes legal perspectives on LLMs, focusing on two main areas: application issues and regulatory frameworks. The first part discusses risks, safety, discrimination, toxicity, and hallucination in LLMs. The second part examines the regulation of large models, including paradigm shifts, adaptations in legal fields, and systemic risks. Additionally, the legal profession must balance ethical considerations with technological use to ensure proper integration, as illustrated in Figure 5, setting the stage for discussions on human-LLM collaboration.

5.1 Technological Ethics in the Application of LLMs

Ethical and human alignment evaluations are crucial for legal AI. Guilherme F.C.F. Almeida et al.'s research found that the consistency between LLMs and human responses in moral and legal reasoning fluctuates, emphasizing that evaluation frameworks need to incorporate contextual factors to enhance credibility [3]. This direction highlights the ethical dimension of legal AI systems and provides key insights for developing more humanized models.

5.1.1 Risks, Safety, and Discrimination Issues of LLMs. In recent years, the rapid development and application of LLMs have prompted reflections on legal, ethical, and social risks. Blodgett et al.

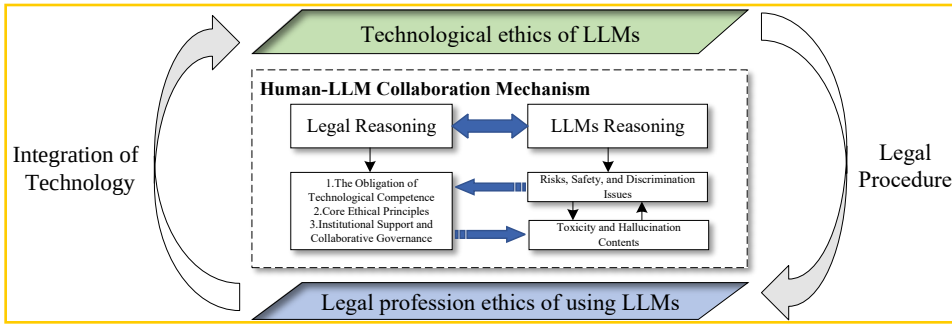


Fig. 6. The Collaboration of Technological Ethics and Legal Ethics when applying LLMs in legal domain

analyzed bias in NLP systems from theoretical and technical angles. Their 2020 study noted that quantitative research on bias often lacks clear motivation and normative reasoning, emphasizing the need to connect language analysis with social power structures to understand technology's harm to specific groups [22]. In 2021, they critiqued fairness benchmark datasets for issues like ambiguous definitions, unverified assumptions, and inconsistent standards [24], proposing the integration of social science methods to redefine measurement frameworks. These insights provide a strong foundation for understanding technological bias.

Weidinger et al. introduced a six-dimensional classification system to expand ethical evaluation in risk frameworks. This system examines how language models may exacerbate social inequality through mechanisms like stereotype reinforcement, unequal resource distribution, and performance disparities. The authors also warned of derivative risks such as misinformation and privacy violations [193]. Building on this, Bommasani et al. analyzed homogenization risks in foundational models like BERT and GPT-3. They emphasized that these models' multitask adaptability could propagate latent flaws into downstream applications, such as legal decision-making systems. Furthermore, the opacity of model outputs may lead to errors, especially in contexts where fairness verification is inadequate, potentially amplifying technical issues at an institutional level [28].

In risk governance, scholars propose complementary solutions. The Bommasani team calls for interdisciplinary cooperation to break the "black box" of models and establish legal interpretability standards. Bender et al. focus on data sources, noting that unfiltered internet data in large models like GPT-3 internalizes biases. They recommend data cleaning and metadata recording to disrupt discriminatory content dissemination, crucial for sensitive applications like legal text generation [15]. Both emphasize ethical integration throughout technology development.

These studies establish a logical framework from analyzing phenomena to governance strategies. Blodgett highlighted methodological flaws in bias research, Weidinger developed a framework for risk perception, and Bommasani and Bender proposed risk mitigation approaches from model architecture and data dimensions. Collectively, they emphasize aligning language model development with legal and ethical frameworks. Only through interdisciplinary collaboration, normative standards, and technological transparency can we balance technological innovation and social equity.

5.1.2 Toxicity and Hallucination Contents of LLMs. When examining the impact of large language models' toxic or hallucinatory outputs on the legal industry, we must address potential biases in processing specific group languages [198]. Blodgett et al. found that NLP tools are less accurate in identifying African American English (AAE) compared to standard English [23, 25]. They suggested using mixed membership models to improve accuracy. These findings highlight NLP biases, which

are critical for ensuring fairness and accuracy in legal texts. Additionally, Weidinger et al. identified six ethical and social risks of large language models, such as discrimination and misinformation, and proposed mitigation strategies emphasizing multi-party collaboration [193]. These studies offer insights into reducing bias in NLP tools for the legal field.

Furthermore, the risks of language models spreading false information are critical concerns requiring legal attention. Zellers et al. introduced the Grover model, designed for generating and detecting neural fake news, emphasizing the importance of releasing robust generators publicly to enhance transparency and accountability [207]. This research highlights the capabilities of language models in producing misleading content. Kreps et al. conducted three experiments examining the credibility of AI-generated text and its impact on public opinion, revealing that people often struggle to distinguish AI-generated from human-written news, with partisan bias affecting credibility assessments but minimally influencing policy perspectives [95]. Buchanan et al. explored GPT-3's potential for spreading false information, assessing its performance across disinformation tasks and demonstrating its ability to generate persuasive content with human assistance [30]. They outlined risks and proposed countermeasures for automated disinformation. Collectively, these findings warn the legal profession: the spread of false information could undermine judicial impartiality and erode public trust in the system.

Finally, detoxifying language models is a challenge the legal industry must consider. Gehman et al. developed the RealToxicityPrompts dataset to evaluate toxic content generation and analyze its sources in training data [64]. Welbl et al. found that while automatic detoxification methods reduce toxicity, they may overfilter marginalized groups and increase bias [194]. These studies offer insights for reducing toxicity in legal texts and ensuring fairness. Additionally, Gorwa et al. examined the technical and political challenges of algorithmic content moderation, including applications in copyright, terrorism, and toxic speech, emphasizing transparency, fairness, and accountability [67].

These findings help the legal industry implement fair and unbiased moderation. LLMs can generate toxic or hallucinatory content, leading to issues such as discrimination against specific groups, spreading false information, and detoxification challenges. These studies offer valuable insights to ensure fairness and accuracy in legal texts while reducing bias.

5.2 The Legal Profession Ethics of Legal Professions Using LLMs

The rapid adoption of LLMs in the legal field has reshaped service delivery, efficiency, and posed challenges to Legal Professional Ethics (LPE). Traditional LPE, centered on obligations like competence, confidentiality, loyalty, diligence, and client communication, requires adaptation when addressing the risks and capabilities of LLMs. This section outlines the ethical guidelines for legal professionals using LLMs, focusing on establishing the "Obligation of Technological Competence" while ensuring other core ethical principles remain intact¹.

5.2.1 The Obligation of Technological Competence: The Core Pillar of Lawyer Ethics in the LLM Era.

The "Obligation of Technological Competence" constitutes a core extension of the ethical duty of lawyers in the LLM era, evolving from the requirement of the American Bar Association (ABA) to understand the benefits and risks of technology [11]. This obligation mandates that lawyers

¹The guidance principles issued by the Bar Council in January 2024 warned against blindly trusting the output of large language models and emphasized the risks of professional negligence, defamation, and data protection claims that may arise from improper use. In November 2023, the Bar Standards Board's guidance principles in the "Risk Outlook Report" pointed out that AI language models are prone to errors because they lack a concept of "reality" and may produce highly plausible but incorrect results. For specific cases, reference may be High Court of Justice, [The Avinide case], [2025] EWHC [1383] (QB).

effectively utilize LLMs while maintaining critical oversight, requiring fundamental understanding of their principles, limitations, and task applicability [115, 138], coupled with proactive risk management of confidentiality breaches, inherent biases, and ethical risks [33, 206].

Central to this duty is the rigorous supervision and verification of LLM outputs, as lawyers retain ultimate responsibility for all work product and must ensure accuracy, legal compliance, and protection of intellectual property [138]. Additionally, the rapidly evolving landscape imposes a continuous learning imperative—through integrated ethics training and CLE—to maintain competence, justify fees, prevent malpractice, and counter de-professionalization pressures [20].

5.2.2 Upholding Core Ethical Principles: Confidentiality, Communication, Loyalty, and Diligence. LLM-assisted legal practice introduces critical ethical obligations requiring strict adherence: (1) Mitigate confidentiality risks by evaluating third-party LLM providers' data policies, prioritizing secure deployment environments, inputting minimally redacted information, and obtaining client consent after risk disclosure [50, 178]; (2) Secure informed consent from clients regarding LLM usage scope, benefits, and confidentiality risks, ensuring transparency [174]. (3) Uphold loyalty/diligence obligations by avoiding over-reliance that compromises personalized analysis or verification rigor; submitting unverified LLM outputs constitutes negligence [1]; (4) Prevent misrepresentation by meticulously verifying against primary sources to eliminate "hallucinated" precedents or plagiarized content, thus avoiding sanctions for misleading submissions [33]. These measures safeguard professional integrity amid technological adoption.

AI-generated legal arguments sometimes reference fictitious cases due to the system's failure to authenticate citations. Legal practitioners must verify AI-provided information against reputable sources. In some cases, attorneys are reminded of their responsibility to ensure the authenticity and originality of LLM-generated content ².

5.2.3 Institutional Support and Collaborative Governance: The Key Roles of Law Firms and Bar Associations. Institutional support from law firms and bar associations is essential for ensuring ethical LLM implementation and upholding attorneys' technological competence obligations. Law firms must establish comprehensive governance frameworks—including approved-use policies, dedicated ethics oversight committees, and rigorous vendor/product assessments for security, compliance, and bias mitigation—while providing mandatory training on LLM risks, verification techniques, and ethical decision-making [143]. Additionally, firms must implement secure infrastructure, standardized client communication protocols (disclosures/consent), and external partnership safeguards to prevent unauthorized practice.

Bar associations play a critical standard-setting role by formally incorporating technological competence into professional conduct rules (e.g., ABA Model Rule 1.1) [4], developing specialized CLE programs on legal tech ethics [9], and bridging the digital divide through equitable resource allocation. Proactive committee structures (e.g., ABA's historical tech ethics initiatives [94], Shanghai's specialized committees [157]) enable authoritative guidance development, enforcement mechanisms for violations, and maintenance of centralized knowledge repositories to ensure uniform ethical standards and service quality across the profession.

6 Roadmap and Future Directions

Based on the comprehensive survey, future research directions for LLMs in law converge on three interconnected pillars:

- **Enhanced Legal Reasoning, Knowledge Integration & Interpretability:** Future work requires multimodal fusion (text, images, charts) to deepen contextual analysis of legal evidence,

²See *R (Ayinde) v London Borough of Haringey*; *Al-Haroun v Qatar National Bank* [2025] EWHC 1383 (Admin)

alongside cross-jurisdictional adaptation through transfer learning for global deployment to reduce hallucinations and inconsistencies. Integrating structured legal knowledge via architectures and structure combining logical rules with neuro-symbolic computation [192] will reduce hallucinations while improving reasoning fidelity. Frameworks grounded in formal argumentation models (for example, Toulmin's warrant-backing structures) will drive accuracy insuch asks such as judgment prediction. Benchmarks such as JUREX-4E [108], LawBench [56] and LegalBench [73] will standardize the the evaluation, particularly for long-document processing and low-resource scenarios. Enhancing the interpretability of a model is also an important research direction. The development of more transparent and interpretable LLM models can help legal professionals better understand and trust the model's output. For example, visualizing the model's reasoning process or developing interpretive algorithms to extract key information can effectively increase the model's transparency and credibility. Optimizing long-document processing capabilities is also an important aspect of technical improvement. By improving methods for handling long documents, such as hierarchical encoding and clustering methods, or developing more efficient long-text segmentation techniques, the model's performance in processing complex legal documents can be significantly enhanced.

- Multi-Agent Workflow Augment:** By developing smarter legal question-answering systems, especially muti-agent workflow, more accurate and comprehensive legal consultations can be provided. For example, through multilingual support and legal knowledge enhancement, the system's performance in different languages and fields can be significantly improved, allowing it to better serve global users [12]. Optimizing legal document processing is also an important practical direction. By optimizing tasks such as the generation, summarization, and classification of legal documents, the efficiency and quality of legal document processing can be significantly increased. For example, through multitask learning and contrastive learning, the model's performance in long-document classification and summarization tasks can be enhanced, thereby better meeting the needs of legal practice. Enhancing legal reasoning and judgment prediction is equally crucial. By combining legal knowledge and case data, the model's performance in legal reasoning and judgment prediction tasks can be significantly improved.
- Ethical & Regulatory Co-Evolution:** Critical In the realm of legal ethics and social impact, reducing bias and discrimination is an important research direction. Through data cleaning, model optimization, and ethical review, bias and discrimination in the application of LLMs in the legal field can be effectively reduced. For example, through fairness analysis and bias detection technologies, the model's fairness and impartiality can be significantly enhanced, thereby better serving society. Enhancing transparency and interpretability is also an important ethical goal. By developing more transparent and interpretable models, legal professionals can better understand and trust the model's output. Establishing a legal ethics framework is also an important direction for development. Through interdisciplinary cooperation, ethical guidelines and regulatory frameworks can be formulated to effectively regulate the application of LLMs in the legal field, ensuring that technological development complies with legal and social ethical requirements. The argumentation framework of the legal profession based on the Toulmin model and its interaction with LLMs can help the legal profession focus more on the functions and goals of legal language [164], thereby fulfilling the professional ethical requirements for the technical competence of the legal profession.

7 Conclusion

The present work synthesizes the integration of LLMs into legal systems through an innovative dual-lens framework combining Toulmin argumentation structures and legal professional roles. It traces the field's evolution from early symbolic AI and task-specific models to contemporary transformer-based LLMs, highlighting breakthroughs in contextual scalability, knowledge integration, and rigorous evaluation benchmarks, which are comprehensively shown in Table 2 and Table 3. The survey systematically categorizes advances across legal reasoning, dispute resolution workflows (litigation and alternative procedures), and ethical governance. Despite progress, critical challenges persist, including hallucination in legal claims, jurisdictional adaptation gaps in low-resource systems, explainability deficits in black-box reasoning, and ethical asymmetries in access and bias. Future work may prioritize multimodal evidence integration, dynamic rebuttal handling, and interdisciplinary frameworks aligning technical innovation with jurisprudential principles to realize robust, ethically grounded legal AI. To address the fundamental tension between algorithmic efficiency and judicial authority—a core controversy previously underexplored—we advocate for a legal-professional-centric strategy. This requires positioning LLMs as assistive tools rather than decision-makers, ensuring human oversight at critical junctures. Technical solutions must prioritize explainable reasoning paths. By centering legal expertise as the ultimate arbiter, we transform the "erosion" debate into a collaborative evolution: one where LLMs augment judicial wisdom while preserving the sacrosanct nature of legal authority.

Acknowledgment

This research was supported by the Humanities and Social Sciences Research Project of the Ministry of Education of China (Grant No. 22YJC820054) and the National Social Science Fund of China (Grant No. 23BZX113).

References

- [1] Andrew Abbott. 1981. Status and status strain in the professions. *American journal of sociology* 86, 4 (1981), 819–835.
- [2] Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altenschmidt, Sam Altman, Shyamal Anadkat, et al. 2023. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774* (2023).
- [3] Guilherme FCF Almeida, José Luiz Nunes, Neele Engelmann, Alex Wiegmann, and Marcelo de Araújo. 2024. Exploring the psychology of LLMs' moral and legal reasoning. *Artificial Intelligence* 333 (2024), 104145.
- [4] American Bar Association Task Force on Law and Artificial Intelligence. 2024. Year 1 Report on the Impact of Artificial Intelligence on the Practice of Law: Legal Challenges. <https://www.americanbar.org/news/abanews/aba-news-archives/2024/08/aba-task-force-report-ai-opportunities-challenges/>. Last visited June 10, 2025.
- [5] Deepa Anand and Rupali Wagh. 2022. Effective deep learning approaches for summarization of legal texts. *Journal of King Saud University-Computer and Information Sciences* 34, 5 (2022), 2141–2150.
- [6] Dang Hoang Anh, Dinh-Truong Do, Vu Tran, and Nguyen Le Minh. 2023. The impact of large language modeling on natural language processing in legal texts: a comprehensive survey. In *2023 15th International Conference on Knowledge and Systems Engineering (KSE)*. IEEE, 1–7.
- [7] Michał Araszkiewicz, Trevor Bench-Capon, Enrico Francesconi, Marc Lauritsen, and Antonino Rotolo. 2022. Thirty years of Artificial Intelligence and Law: overviews. *Artificial Intelligence and Law* 30, 4 (2022), 593–610.
- [8] Farid Ariai and Gianluca Demartini. 2024. Natural Language Processing for the Legal Domain: A Survey of Tasks, Datasets, Models, and Challenges. *arXiv preprint arXiv:2410.21306* (2024).
- [9] A. Arrington. 2024. "Purposefully Vague" or Problematic? Why Lawyers Must Define the Duty of Tech Competence. *University of St. Thomas Law Journal* 20 (2024), 218–245.
- [10] Kevin D Ashley. 2017. *Artificial intelligence and legal analytics: new tools for law practice in the digital age*. Cambridge University Press.
- [11] Jamie J Baker. 2017. Beyond the information age: the duty of technology competence in the algorithmic society. *SCL Rev* 69 (2017), 557.
- [12] Ryan C Barron, Maksim E Eren, Olga M Serafimova, Cynthia Matuszek, and Boian S Alexandrov. 2025. Bridging Legal Knowledge and AI: Retrieval-Augmented Generation with Vector Stores, Knowledge Graphs, and Hierarchical

Non-negative Matrix Factorization. *arXiv preprint arXiv:2502.20364* (2025).

- [13] Mihir Abhay Bedekar, Manoj Pareek, Saurabh Suman Choudhuri, PF Abhishek, Jayesh Jhurani, and Rutul Shah. 2024. AI in Mergers and Acquisitions: Analyzing the Effectiveness of Artificial Intelligence in Due Diligence. In *2024 International Conference on Knowledge Engineering and Communication Systems (ICKECS)*, Vol. 1. IEEE, 1–5.
- [14] Iz Beltagy, Matthew E Peters, and Arman Cohan. 2020. Longformer: The long-document transformer. *arXiv preprint arXiv:2004.05150* (2020).
- [15] Emily M Bender, Timnit Gebru, Angelina McMillan-Major, and Shmargaret Shmitchell. 2021. On the dangers of stochastic parrots: Can language models be too big?. In *Proceedings of the 2021 ACM conference on fairness, accountability, and transparency*. 610–623.
- [16] Irene Benedetto, Luca Cagliero, Michele Ferro, Francesco Tarasconi, Claudia Bernini, and Giuseppe Giacalone. 2025. Leveraging large language models for abstractive summarization of Italian legal news. *Artificial Intelligence and Law* (2025), 1–21.
- [17] Floris Bex, Henry Prakken, Chris Reed, and Douglas Walton. 2003. Towards a formal account of reasoning about evidence: argumentation schemes and generalisations. *Artificial Intelligence and Law* 11 (2003), 125–165.
- [18] Floris Bex and Bart Verheij. 2013. Legal stories and the process of proof. *Artificial Intelligence and Law* 21 (2013), 253–278.
- [19] Floris J Bex. 2011. *Arguments, stories and criminal evidence: A formal hybrid theory*. Vol. 92. Springer Science & Business Media.
- [20] Piotr Bialowolski and Dorota Weziak-Bialowolska. 2021. What Does It Take to Be a Good Lawyer? The Underpinnings of Success in a Rapidly Growing Legal Market. *Sustainability* 13, 11 (2021), 5841.
- [21] Onur Bilgin, Logan Fields, Antonio Laverghetta Jr, Zaid Marji, Animesh Nighojkar, Stephen Steinle, and John Licato. 2024. Exploring Prompting Approaches in Legal Textual Entailment. *The Review of Socionetwork Strategies* (2024), 1–26.
- [22] Su Lin Blodgett, Solon Barocas, Hal Daumé III, and Hanna Wallach. 2020. Language (technology) is power: A critical survey of “bias” in nlp. *arXiv preprint arXiv:2005.14050* (2020).
- [23] Su Lin Blodgett, Lisa Green, and Brendan O’Connor. 2016. Demographic dialectal variation in social media: A case study of African-American English. *arXiv preprint arXiv:1608.08868* (2016).
- [24] Su Lin Blodgett, Gilsinia Lopez, Alexandra Olteanu, Robert Sim, and Hanna Wallach. 2021. Stereotyping Norwegian salmon: An inventory of pitfalls in fairness benchmark datasets. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*. 1004–1015.
- [25] Su Lin Blodgett and Brendan O’Connor. 2017. Racial disparity in natural language processing: A case study of social media african-american english. *arXiv preprint arXiv:1707.00061* (2017).
- [26] Guido Boella, Luigi Di Caro, and Llio Humphreys. 2011. Using classification to support legal knowledge engineers in the Eunomos legal document management system. In *Fifth international workshop on juris-informatics (JURISIN)*.
- [27] Michael Bommarito II and Daniel Martin Katz. 2022. GPT takes the bar exam. *arXiv preprint arXiv:2212.14402* (2022).
- [28] Rishi Bommasani, Drew A Hudson, Ehsan Adeli, Russ Altman, Simran Arora, Sydney von Arx, Michael S Bernstein, Jeannette Bohg, Antoine Bosselut, Emma Brunskill, et al. 2021. On the opportunities and risks of foundation models. *arXiv preprint arXiv:2108.07258* (2021).
- [29] Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. 2020. Language models are few-shot learners. *Advances in neural information processing systems* 33 (2020), 1877–1901.
- [30] Ben Buchanan, Andrew Lohn, Micah Musser, and Katerina Sedova. 2021. Truth, lies, and automation. *Center for Security and Emerging technology* 1, 1 (2021), 2.
- [31] Minh-Quan Bui, Dinh-Truong Do, Nguyen-Khang Le, Dieu-Hien Nguyen, Khac-Vu-Hiep Nguyen, Trang Pham Ngoc Anh, and Minh Le Nguyen. 2024. Data Augmentation and Large Language Model for Legal Case Retrieval and Entailment. *The Review of Socionetwork Strategies* 18, 1 (2024), 49–74.
- [32] Roberta Calegari, Giuseppe Contissa, Francesca Lagioia, Andrea Omicini, and Giovanni Sartor. 2019. Defeasible systems in legal reasoning: A comparative assessment. In *Legal Knowledge and Information Systems*. IOS Press, 169–174.
- [33] Salvatore Caserta. 2022. The sociology of the legal profession in the digital age. *International Journal of the Legal Profession* 29, 3 (2022), 319–334.
- [34] Ilias Chalkidis and Ion Androutsopoulos. 2017. A deep learning approach to contract element extraction. In *Legal knowledge and information systems*. IOS Press, 155–164.
- [35] Ilias Chalkidis, Ion Androutsopoulos, and Nikolaos Aletras. 2019. Neural legal judgment prediction in English. *arXiv preprint arXiv:1906.02059* (2019).

- [36] Ilias Chalkidis, Abhik Jana, Dirk Hartung, Michael Bommarito, Ion Androutsopoulos, Daniel Martin Katz, and Nikolaos Aletras. 2021. LexGLUE: A benchmark dataset for legal language understanding in English. *arXiv preprint arXiv:2110.00976* (2021).
- [37] Haihua Chen, Lei Wu, Jiangping Chen, Wei Lu, and Junhua Ding. 2022. A comparative study of automated legal text classification using random forests and deep learning. *Information Processing & Management* 59, 2 (2022), 102798.
- [38] Kevin Clark, Minh-Thang Luong, Quoc V Le, and Christopher D Manning. 2020. Electra: Pre-training text encoders as discriminators rather than generators. *arXiv preprint arXiv:2003.10555* (2020).
- [39] Joe Collenette, Katie Atkinson, and Trevor Bench-Capon. 2023. Explainable AI tools for legal reasoning about cases: A study on the European Court of Human Rights. *Artificial Intelligence* 317 (2023), 103861.
- [40] Florin Cuconasu, Simone Filice, Guy Horowitz, Yoelle Maarek, and Fabrizio Silvestri. 2025. Do RAG Systems Suffer From Positional Bias? *arXiv preprint arXiv:2505.15561* (2025).
- [41] Jiayi Cui, Zongjian Li, Yang Yan, Bohua Chen, and Li Yuan. 2023. Chatlaw: Open-source legal large language model with integrated external knowledge bases. *CoRR* (2023).
- [42] Jiayi Cui, Munan Ning, Zongjian Li, Bohua Chen, Yang Yan, Hao Li, Bin Ling, Yonghong Tian, and Li Yuan. 2024. Chatlaw: A multi-agent collaborative legal assistant with knowledge graph enhanced mixture-of-experts large language model. *arXiv preprint arXiv:2306.16092* (2024).
- [43] Junyun Cui, Xiaoyu Shen, and Shaochun Wen. 2023. A survey on legal judgment prediction: Datasets, metrics, models and challenges. *IEEE Access* (2023).
- [44] Matthew Dahl, Varun Magesh, Mirac Suzgun, and Daniel E Ho. 2024. Large legal fictions: Profiling legal hallucinations in large language models. *Journal of Legal Analysis* 16, 1 (2024), 64–93.
- [45] Yongfu Dai, DuanYu Feng, Jimin Huang, Haochen Jia, Qianqian Xie, Yifang Zhang, Weiguang Han, Wei Tian, and Hao Wang. 2023. LAIW: a Chinese legal large language models benchmark. *arXiv preprint arXiv:2310.05620* (2023).
- [46] Chenlong Deng, Kelong Mao, Yuyao Zhang, and Zhicheng Dou. 2024. Enabling discriminative reasoning in llms for legal judgment prediction. *arXiv preprint arXiv:2407.01964* (2024).
- [47] Wentao Deng, Jiahuan Pei, Keyi Kong, Zhe Chen, Furu Wei, Yujun Li, Zhaochun Ren, Zhumin Chen, and Pengjie Ren. 2023. Sylogistic reasoning for legal judgment analysis. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*. 13997–14009.
- [48] Aniket Deroy, Kripabandhu Ghosh, and Saptarshi Ghosh. 2024. Applicability of large language models and generative models for legal case judgement summarization. *Artificial Intelligence and Law* (2024), 1–44.
- [49] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, volume 1 (long and short papers)*. 4171–4186.
- [50] Sanjeev Dewan and Frederick J Riggins. 2005. The digital divide: Current and future research directions. *Journal of the Association for information systems* 6, 12 (2005), 298–337.
- [51] Johannes Dimyadi, Guido Governatori, and Robert Amor. 2017. Evaluating legaldocml and legalruleml as a standard for sharing normative information in the aec/im domain. In *Proceedings of the Joint Conference on Computing in Construction (7C3)*, Vol. 1. Heriot-Watt University, Edinburgh, UK, Heraklion, Greece, 637–644.
- [52] Stefan Daniel Dumitrescu and Andrei-Marius Avram. 2019. Introducing RONEC—the Romanian Named Entity Corpus. *arXiv preprint arXiv:1909.01247* (2019).
- [53] Ronald Dworkin. 1986. *Law's empire*. Harvard University Press.
- [54] Mohamed Elaraby, Huihui Xu, Morgan Gray, Kevin D Ashley, and Diane Litman. 2024. Adding Argumentation into Human Evaluation of Long Document Abstractive Summarization: A Case Study on Legal Opinions. In *Proceedings of the Fourth Workshop on Human Evaluation of NLP Systems (HumEval)@ LREC-COLING 2024*. 28–35.
- [55] Yu Fan, Jingwei Ni, Jakob Merane, Etienne Salimbeni, Yang Tian, Yoan Hermstrüwer, Yinya Huang, Mubashara Akhtar, Florian Geering, Oliver Dreyer, et al. 2025. LEXam: Benchmarking Legal Reasoning on 340 Law Exams. *arXiv preprint arXiv:2505.12864* (2025).
- [56] Zhiwei Fei, Xiaoyu Shen, Dawei Zhu, Fengzhe Zhou, Zhuo Han, Songyang Zhang, Kai Chen, Zongwen Shen, and Jidong Ge. 2023. Lawbench: Benchmarking legal knowledge of large language models. *arXiv preprint arXiv:2309.16289* (2023).
- [57] Zhiwei Fei, Songyang Zhang, Xiaoyu Shen, Dawei Zhu, Xiao Wang, Maosong Cao, Fengzhe Zhou, Yining Li, Wenwei Zhang, Dahua Lin, et al. 2024. Internlm-law: An open source chinese legal large language model. *arXiv preprint arXiv:2406.14887* (2024).
- [58] Yi Feng, Chuanyi Li, and Vincent Ng. 2022. Legal judgment prediction via event extraction with constraints. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. 648–664.
- [59] Enrico Francesconi. 2014. A description logic framework for advanced accessing and reasoning over normative provisions. *Artificial intelligence and Law* 22, 3 (2014), 291–311.

- [60] Masaki Fujita, Takaaki Onaga, and Yoshinobu Kano. 2024. LLM Tuning and Interpretable CoT: KIS Team in COLIEE 2024. In *JSAI International Symposium on Artificial Intelligence*. Springer, 140–155.
- [61] Leilei Gan, Kun Kuang, Yi Yang, and Fei Wu. 2021. Judgment prediction via injecting legal knowledge into neural networks. In *Proceedings of the AAAI conference on artificial intelligence*, Vol. 35. 12866–12874.
- [62] I García Fernández. 2024. *Ontology Engineering with Large Language Models*. (2024).
- [63] Jidong Ge, Yunyun Huang, Xiaoyu Shen, Chuanyi Li, and Wei Hu. 2021. Learning fine-grained fact-article correspondence in legal cases. *IEEE/ACM Transactions on Audio, Speech, and Language Processing* 29 (2021), 3694–3706.
- [64] Samuel Gehman, Suchin Gururangan, Maarten Sap, Yejin Choi, and Noah A Smith. 2020. Realtotoxicityprompts: Evaluating neural toxic degeneration in language models. *arXiv preprint arXiv:2009.11462* (2020).
- [65] Sudipto Ghosh, Devanshu Verma, Balaji Ganesan, Purnima Bindal, Vikas Kumar, and Vasudha Bhatnagar. 2024. Human Centered AI for Indian Legal Text Analytics. *arXiv preprint arXiv:2403.10944* (2024).
- [66] Randy Goebel, Yoshinobu Kano, Mi-Young Kim, Julian Kabele, Ken Satoh, and Masaharu Yoshioka. 2024. Overview and Discussion of the Competition on Legal Information, Extraction/Entailment (COLIEE) 2023. *The Review of Socionetwork Strategies* 18, 1 (2024), 27–47.
- [67] Robert Gorwa, Reuben Binns, and Christian Katzenbach. 2020. Algorithmic content moderation: Technical and political challenges in the automation of platform governance. *Big Data & Society* 7, 1 (2020), 2053951719897945.
- [68] Shreya Goswami, Naveen Saini, and Saurabh Shukla. 2025. Incorporating Domain Knowledge in Multi-objective Optimization Framework for Automating Indian Legal Case Summarization. In *International Conference on Pattern Recognition*. Springer, 265–280.
- [69] S Georgette Graham, Hamidreza Soltani, and Olufemi Isiaq. 2023. Natural language processing for legal document review: categorising deontic modalities in contracts. *Artificial Intelligence and Law* (2023), 1–22.
- [70] Morgan A Gray, Jaromir Savelka, Wesley M Oliver, and Kevin D Ashley. 2024. Empirical legal analysis simplified: reducing complexity through automatic identification and evaluation of legally relevant factors. *Philosophical Transactions of the Royal Society A* 382, 2270 (2024), 20230155.
- [71] Paul W Grimm. 2017. Challenges facing judges regarding expert evidence in criminal cases. *Fordham L. Rev.* 86 (2017), 1601.
- [72] Thomas R Gruber. 1991. The role of common ontology in achieving sharable, reusable knowledge bases. *Kr* 91 (1991), 601–602.
- [73] Neel Guha, Julian Nyarko, Daniel Ho, Christopher Ré, Adam Chilton, Alex Chohlas-Wood, Austin Peters, Brandon Waldon, Daniel Rockmore, Diego Zambrano, et al. 2024. Legalbench: A collaboratively built benchmark for measuring legal reasoning in large language models. *Advances in Neural Information Processing Systems* 36 (2024).
- [74] Herbert Lionel Adolphus Hart and Leslie Green. 2012. *The concept of law*. oxford university press.
- [75] Congqing He, Tien-Ping Tan, Sheng Xue, and Yanyu Tan. 2025. Simulating judicial trial logic: Dual residual cross-attention learning for predicting legal judgment in long documents. *Expert Systems with Applications* 261 (2025), 125462.
- [76] Amanda Head and Sonya Willis. 2024. Assessing law students in a GenAI world to create knowledgeable future lawyers. *International Journal of the Legal Profession* 31, 3 (2024), 293–310.
- [77] Dan Hendrycks, Collin Burns, Anya Chen, and Spencer Ball. 2021. CUAD: an expert-annotated NLP dataset for legal contract review. *arXiv preprint arXiv:2103.06268* (2021).
- [78] Felix Hill, Kyunghyun Cho, Anna Korhonen, and Yoshua Bengio. 2016. Learning to understand phrases by embedding the dictionary. *Transactions of the Association for Computational Linguistics* 4 (2016), 17–30.
- [79] David Hitchcock. 2005. Good reasoning on the Toulmin model. *Argumentation* 19 (2005), 373–391.
- [80] Wesley Newcomb Hohfeld. 1917. Fundamental legal conceptions as applied in judicial reasoning. *The Yale Law Journal* 26, 8 (1917), 710–770.
- [81] John Hokkanen and Marc Lauritsen. 2002. Knowledge tools for legal knowledge tool makers. *Artificial Intelligence and Law* 10, 4 (2002), 295–302.
- [82] Yin Hua and Wu Zihao. 2024. Mixture of Expert Large Language Model for Legal Case Element Recognition. *Journal of Frontiers of Computer Science and Technology* 18, 12 (2024), 3260–3271.
- [83] Quzhe Huang, Mingxu Tao, Chen Zhang, Zhenwei An, Cong Jiang, Zhibin Chen, Zirui Wu, and Yansong Feng. 2023. Lawyer llama technical report. *arXiv preprint arXiv:2305.15062* (2023).
- [84] Yunyun Huang, Xiaoyu Shen, Chuanyi Li, Jidong Ge, and Bin Luo. 2021. Dependency learning for legal judgment prediction with a unified text-to-text transformer. *arXiv preprint arXiv:2112.06370* (2021).
- [85] John Hudzina, Kanika Madan, Dhivya Chinnappa, Jinane Harmouche, Hiroko Bretz, Andrew Vold, and Frank Schilder. 2020. Information extraction/entailment of common law and civil code. In *JSAI International Symposium on Artificial Intelligence*. Springer, 254–268.
- [86] Wonseok Hwang, Dongjun Lee, Kyoungyeon Cho, Hanuul Lee, and Minjoon Seo. 2022. A multi-task benchmark for korean legal language understanding and judgement prediction. *Advances in Neural Information Processing Systems*

- 35 (2022), 32537–32551.
- [87] Ali Shariq Imran, Henrik Hodnefeld, Zenun Kastrati, Noureen Fatima, Sher Muhammad Daudpota, and Mudasir Ahmad Wani. 2023. Classifying European court of human rights cases using transformer-based techniques. *IEEE Access* 11 (2023), 55664–55676.
- [88] Deepali Jain, Malaya Dutta Borah, and Anupam Biswas. 2024. Domain knowledge-enriched summarization of legal judgment documents via grey wolf optimization. In *Advances in Computers*. Vol. 135. Elsevier, 233–258.
- [89] Deepali Jain, Malaya Dutta Borah, and Anupam Biswas. 2024. Summarization of Lengthy Legal Documents via Abstractive Dataset Building: An Extract-then-Assign Approach. *Expert Systems with Applications* 237 (2024), 121571.
- [90] Mi-Young Kim, Juliano Rabelo, Kingsley Okeke, and Randy Goebel. 2022. Legal information retrieval and entailment based on bm25, transformer and semantic thesaurus methods. *The Review of Socionetwork Strategies* 16, 1 (2022), 157–174.
- [91] Ryan Kiros, Yukun Zhu, Russ R Salakhutdinov, Richard Zemel, Raquel Urtasun, Antonio Torralba, and Sanja Fidler. 2015. Skip-thought vectors. *Advances in neural information processing systems* 28 (2015).
- [92] Wolters Kluwer. 2021. The 2021 Wolters Kluwer Future Ready Lawyer. Moving Beyond the Pandemic. Survey Report.
- [93] Charles W Kneupper. 1978. Teaching argument: An introduction to the Toulmin model. *College Composition & Communication* 29, 3 (1978), 237–241.
- [94] T. V. Kolb. 2016. Technology Competence: The New Ethical Mandate for North Dakota Lawyers and the Practice of Law. *North Dakota Law Review* 92 (2016), 91–113.
- [95] Sarah Kreps, R Miles McCain, and Miles Brundage. 2022. All the news that's fit to fabricate: AI-generated text as a tool of media misinformation. *Journal of experimental political science* 9, 1 (2022), 104–117.
- [96] Jinqi Lai, Wensheng Gan, Jiayang Wu, Zhenlian Qi, and Philip S Yu. 2024. Large language models in law: A survey. *AI Open* (2024).
- [97] Zhenzhong Lan, Mingda Chen, Sebastian Goodman, Kevin Gimpel, Piyush Sharma, and Radu Soricut. 2019. Albert: A lite bert for self-supervised learning of language representations. *arXiv preprint arXiv:1909.11942* (2019).
- [98] Marc Lauritsen. 1992. Technology report: Building legal practice systems with today's commercial authoring tools. *Artificial Intelligence and Law* 1 (1992), 87–102.
- [99] Marc Lauritsen. 1995. Technology report: Work product retrieval systems in today's law offices. *Artificial Intelligence and Law* 3 (1995), 287–304.
- [100] Valentina Leone, Luigi Di Caro, and Serena Villata. 2020. Taking stock of legal ontologies: a feature-based comparative analysis. *Artificial Intelligence and Law* 28, 2 (2020), 207–235.
- [101] Edward H Levi. 2013. *An introduction to legal reasoning*. University of Chicago Press.
- [102] Mike Lewis, Yinhan Liu, Naman Goyal, Marjan Ghazvininejad, Abdelrahman Mohamed, Omer Levy, Ves Stoyanov, and Luke Zettlemoyer. 2019. Bart: Denoising sequence-to-sequence pre-training for natural language generation, translation, and comprehension. *arXiv preprint arXiv:1910.13461* (2019).
- [103] Haitao Li, You Chen, Zhekai Ge, Qingyao Ai, Yiqun Liu, Quan Zhou, and Shuai Huo. 2024. Towards an In-Depth Comprehension of Case Relevance for Better Legal Retrieval. In *JSAI International Symposium on Artificial Intelligence*. Springer, 212–227.
- [104] Shangyuan Li, Shiman Zhao, Zhuoran Zhang, Zihao Fang, Wei Chen, and Tengjiao Wang. 2025. Basis is also explanation: Interpretable Legal Judgment Reasoning prompted by multi-source knowledge. *Information Processing & Management* 62, 3 (2025), 103996.
- [105] Davide Liga and Livio Robaldo. 2023. Fine-tuning GPT-3 for legal rule classification. *Computer Law & Security Review* 51 (2023), 105864.
- [106] Marco Lippi and Paolo Torroni. 2016. Argumentation mining: State of the art and emerging trends. *ACM Transactions on Internet Technology (TOIT)* 16, 2 (2016), 1–25.
- [107] Dugang Liu, Weihao Du, Lei Li, WeiKe Pan, and Zhong Ming. 2022. Augmenting legal judgment prediction with contrastive case relations. In *Proceedings of the 29th International Conference on Computational Linguistics*. 2658–2667.
- [108] Huanghai Liu, Quzhe Huang, Qingjing Chen, Yiran Hu, Jiayu Ma, Yun Liu, Weixing Shen, and Yansong Feng. 2025. Jurex-4e: Juridical expert-annotated four-element knowledge base for legal reasoning. *arXiv preprint arXiv:2502.17166* (2025).
- [109] Shuaiqi Liu, Jiannong Cao, Yicong Li, Ruosong Yang, and Zhiyuan Wen. 2024. Low-resource court judgment summarization for common law systems. *Information Processing & Management* 61, 5 (2024), 103796.
- [110] Lajanugen Logeswaran, Honglak Lee, and Dragomir Radev. 2018. Sentence ordering and coherence modeling using recurrent neural networks. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 32.
- [111] Yougang Lyu, Zihan Wang, Zhaochun Ren, Pengjie Ren, Zhumin Chen, Xiaozhong Liu, Yujun Li, Hongsong Li, and Hongye Song. 2022. Improving legal judgment prediction through reinforced criminal element extraction. *Information Processing & Management* 59, 1 (2022), 102780.

- [112] Luyao Ma, Wei Ye, and Shikun Zhang. 2020. Judgment Prediction Based on Case Life Cycle. In *The 1st International Workshop on Legal Intelligence Held in conjunction with SIGIR*.
- [113] Luyao Ma, Yating Zhang, Tianyi Wang, Xiaozhong Liu, Wei Ye, Changlong Sun, and Shikun Zhang. 2021. Legal judgment prediction with multi-stage case representation learning in the real court setting. In *Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval*. 993–1002.
- [114] Neil MacCormick. 1994. *Legal reasoning and legal theory*. Clarendon Press.
- [115] Karolina Mania. 2023. Legal technology: assessment of the legal tech industry's potential. *Journal of the Knowledge Economy* 14, 2 (2023), 595–619.
- [116] Qiang Mao, Adam Dabrowski, Fusheng Wei, Eric Olson, Robert Neary, Jingchao Yang, Han Qin, and Nathaniel Huber-Fliflet. 2024. Comparative Analysis of LLM-Generated Event Timeline Summarization for Legal Investigations. In *2024 IEEE International Conference on Big Data (BigData)*. IEEE, 4743–4750.
- [117] Mohammed Maree, Rabee Al-Qasem, and Banan Tantour. 2024. Transforming legal text interactions: leveraging natural language processing and large language models for legal support in Palestinian cooperatives. *International Journal of Information Technology* 16, 1 (2024), 551–558.
- [118] Lauren Martin, Nick Whitehouse, Stephanie Yiu, Lizzie Catterson, and Rivindu Perera. 2024. Better call gpt, comparing large language models against lawyers. *arXiv preprint arXiv:2401.16212* (2024).
- [119] Jorge Martinez-Gil. 2023. A survey on legal question-answering systems. *Computer Science Review* 48 (2023), 100552.
- [120] Masha Medvedeva, Martijn Wieling, and Michel Vols. 2023. Rethinking the field of automatic prediction of court decisions. *Artificial Intelligence and Law* 31, 1 (2023), 195–212.
- [121] Eliza Mik. 2022. Much ado about artificial intelligence or: the automation of contract formation. *International Journal of Law and Information Technology* 30, 4 (2022), 484–506.
- [122] Tomas Mikolov. 2013. Efficient estimation of word representations in vector space. *arXiv preprint arXiv:1301.3781* 3781 (2013).
- [123] Tomas Mikolov, Ilya Sutskever, Kai Chen, Greg S Corrado, and Jeff Dean. 2013. Distributed representations of words and phrases and their compositionality. *Advances in neural information processing systems* 26 (2013).
- [124] Gian Carlo Milanese, Georgios Peikos, Gabriella Pasi, and Marco Viviani. 2025. Fact-Driven Health Information Retrieval: Integrating LLMs and Knowledge Graphs to Combat Misinformation. In *European Conference on Information Retrieval*. Springer, 192–200.
- [125] Gianluca Moro, Nicola Piscaglia, Luca Ragazzi, and Paolo Italiani. 2024. Multi-language transfer learning for low-resource legal case summarization. *Artificial Intelligence and Law* 32, 4 (2024), 1111–1139.
- [126] Laura Nader and Harry F Todd. 1978. *The disputing process—Law in ten societies*. Columbia University Press.
- [127] Chau Nguyen and Le-Minh Nguyen. 2024. Employing label models on ChatGPT answers improves legal text entailment performance. *arXiv preprint arXiv:2401.17897* (2024).
- [128] Chau Nguyen, Phuong Nguyen, and Le-Minh Nguyen. 2025. Retrieve–Revise–Refine: A novel framework for retrieval of concise entailing legal article set. *Information Processing & Management* 62, 1 (2025), 103949.
- [129] Chau Nguyen, Thanh Tran, Khang Le, Hien Nguyen, Truong Do, Trang Pham, Son T Luu, Trung Vo, and Le-Minh Nguyen. 2024. Pushing the boundaries of legal information processing with integration of large language models. In *JSAI International Symposium on Artificial Intelligence*. Springer, 167–182.
- [130] Duy-Hung Nguyen, Bao-Sinh Nguyen, Nguyen Viet Dung Nghiem, Dung Tien Le, Mim Amina Khatun, Minh-Tien Nguyen, and Hung Le. 2021. Robust deep reinforcement learning for extractive legal summarization. In *Neural Information Processing: 28th International Conference, ICONIP 2021, Sanur, Bali, Indonesia, December 8–12, 2021, Proceedings, Part VI* 28. Springer, 597–604.
- [131] Ha-Thanh Nguyen, Minh-Phuong Nguyen, Thi-Hai-Yen Vuong, Minh-Quan Bui, Minh-Chau Nguyen, Tran-Binh Dang, Vu Tran, Le-Minh Nguyen, and Ken Satoh. 2022. Transformer-based approaches for legal text processing: Jnl team-coliee 2021. *The Review of Socionetwork Strategies* 16, 1 (2022), 135–155.
- [132] Ha-Thanh Nguyen, Manh-Kien Phi, Xuan-Bach Ngo, Vu Tran, Le-Minh Nguyen, and Minh-Phuong Tu. 2024. Attentive deep neural networks for legal document retrieval. *Artificial Intelligence and Law* 32, 1 (2024), 57–86.
- [133] Tan-Minh Nguyen, Hai-Long Nguyen, Dieu-Quynh Nguyen, Hoang-Trung Nguyen, Thi-Hai-Yen Vuong, and Ha-Thanh Nguyen. 2024. NOWJ@ COLIEE 2024: leveraging advanced deep learning techniques for efficient and effective legal information processing. In *JSAI International Symposium on Artificial Intelligence*. Springer, 183–199.
- [134] Truong-Son Nguyen, Le-Minh Nguyen, Satoshi Tojo, Ken Satoh, and Akira Shimazu. 2018. Recurrent neural network-based models for recognizing requisite and effectuation parts in legal texts. *Artificial Intelligence and Law* 26 (2018), 169–199.
- [135] Shubham Kumar Nigam, Aniket Deroy, Subhankar Maity, and Arnab Bhattacharya. 2024. Rethinking legal judgement prediction in a realistic scenario in the era of large language models. *arXiv preprint arXiv:2410.10542* (2024).
- [136] Joel Niklaus, Ilias Chalkidis, and Matthias Stürmer. 2021. Swiss-judgment-prediction: A multilingual legal judgment prediction benchmark. *arXiv preprint arXiv:2110.00806* (2021).

- [137] Fife Ogunde. 2024. Navigating the legal landscape: large language models and the hesitancy of legal professionals. *International Journal of the Legal Profession* 31, 3 (2024), 311–322.
- [138] Dyane L O’Leary. 2020. “Smart” Lawyering: Integrating Technology Competence into the Legal Practice Curriculum. *UNHL Rev.* 19 (2020), 197.
- [139] Takaaki Onaga, Masaki Fujita, and Yoshinobu Kano. 2024. Contribution Analysis of Large Language Models and Data Augmentations for Person Names in Solving Legal Bar Examination at COLIEE 2023. *The Review of Socionetwork Strategies* 18, 1 (2024), 123–143.
- [140] Anja Oskamp and Marc Lauritsen. 2002. AI in law practice? So far, not much. *AI & L.* 10 (2002), 227.
- [141] Sumit Pai, Soumik Lahiri, Ujjwal Kumar, Krishanu Bakshi, Elijah Soba, Michael Snesserman, Nirmala Pudota, Jon Foster, Edward Bowen, and Sanmitra Bhattacharya. 2023. Exploration of open large language models for ediscovery. In *Proceedings of the Natural Legal Language Processing Workshop 2023*. 166–177.
- [142] Raquel Mochales Palau and Marie-Francine Moens. 2009. Argumentation mining: the detection, classification and structure of arguments in text. In *Proceedings of the 12th international conference on artificial intelligence and law*. 98–107.
- [143] Nicolás Parra-Herrera. 2024. Being a Competent Lawyer in the Age of Generative Artificial Intelligence: A Student Fellow Project. <https://clp.law.harvard.edu/knowledge-hub/insights/being-a-competent-lawyer-in-the-age-of-generative-artificial-intelligence/>. Harvard Law School Center on the Legal Profession, last visited June 10, 2025.
- [144] Jeffrey Pennington, Richard Socher, and Christopher D Manning. 2014. Glove: Global vectors for word representation. In *Proceedings of the 2014 conference on empirical methods in natural language processing (EMNLP)*. 1532–1543.
- [145] Wim Peters, Maria-Teresa Sagri, and Daniela Tiscornia. 2007. The structuring of legal knowledge in LOIS. *Artificial Intelligence and Law* 15 (2007), 117–135.
- [146] Thiago Dal Pont, Federico Galli, Andrea Loreggia, Giuseppe Pisano, Riccardo Rovatti, and Giovanni Sartor. 2023. Legal Summarisation through LLMs: The PRODIGIT Project. *arXiv preprint arXiv:2308.04416* (2023).
- [147] Nishchal Prasad, Mohand Boughanem, and Taoufiq Dkaki. 2023. IRIT_IRIS_C at SemEval-2023 Task 6: A Multi-level Encoder-based Architecture for Judgement Prediction of Legal Cases and their Explanation. In *Proceedings of the 17th International Workshop on Semantic Evaluation (SemEval-2023)*. 686–692.
- [148] Nishchal Prasad, Mohand Boughanem, and Taoufiq Dkaki. 2024. Exploring Large Language Models and Hierarchical Frameworks for Classification of Large Unstructured Legal Documents. In *European Conference on Information Retrieval*. Springer, 221–237.
- [149] Weicong Qin and Zhongxiang Sun. 2024. Exploring the Nexus of Large Language Models and Legal Systems: A Short Survey. *arXiv preprint arXiv:2404.00990* (2024).
- [150] Juliano Rabelo, Mi-Young Kim, and Randy Goebel. 2019. Combining similarity and transformer methods for case law entailment. In *Proceedings of the seventeenth international conference on artificial intelligence and law*. 290–296.
- [151] Alec Radford, Karthik Narasimhan, Tim Salimans, Ilya Sutskever, et al. 2018. Improving language understanding by generative pre-training. (2018).
- [152] Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, Ilya Sutskever, et al. 2019. Language models are unsupervised multitask learners. *OpenAI blog* 1, 8 (2019), 9.
- [153] Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J Liu. 2020. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of machine learning research* 21, 140 (2020), 1–67.
- [154] Sneha Ann Reji, Reshma Sheik, A Sharon, Avisha Rai, and S Jaya Nirmala. 2024. Enhancing LLM Performance on Legal Textual Entailment with Few-Shot CoT-based RAG. In *2024 IEEE International Conference on Signal Processing, Informatics, Communication and Energy Systems (SPICES)*. IEEE, 1–6.
- [155] Adam Roegiest, Alexander K Hudek, and Anne McNulty. 2018. A dataset and an examination of identifying passages for due diligence. In *The 41st international ACM SIGIR conference on research & development in information retrieval*. 465–474.
- [156] TYS Santosh, Oana Ichim, and Matthias Grabmair. 2023. Zero-shot transfer of article-aware legal outcome classification for european court of human rights cases. *arXiv preprint arXiv:2302.00609* (2023).
- [157] Shanghai Lawyers Association. 2025. Special Committees. <https://www.lawyers.org.cn/aboutus/special-committees>. Last visited June 11, 2025.
- [158] Hsuan-Lei Shao, Yi-Chia Chen, and Sieh-Chuen Huang. 2021. BERT-based ensemble model for statute law retrieval and legal information entailment. In *New Frontiers in Artificial Intelligence: JSAI-isAI 2020 Workshops, JURISIN, LENLS 2020 Workshops, Virtual Event, November 15–17, 2020, Revised Selected Papers 12*. Springer, 226–239.
- [159] Rui Shao, Yiping Tang, Lingyan Yang, and Fang Wang. 2025. Law LLM Unlearning via Interfere Prompt, Review Output and Update Parameter: New Challenges, Method and Baseline. *Expert Systems with Applications* (2025), 128612.

- [160] Yunqiu Shao, Jiaxin Mao, Yiqun Liu, Weizhi Ma, Ken Satoh, Min Zhang, and Shaoping Ma. 2020. BERT-PLI: Modeling paragraph-level interactions for legal case retrieval.. In *IJCAL*. 3501–3507.
- [161] Yunqiu Shao, Yueyue Wu, Yiqun Liu, Jiaxin Mao, and Shaoping Ma. 2023. Understanding relevance judgments in legal case retrieval. *ACM Transactions on Information Systems* 41, 3 (2023), 1–32.
- [162] Zhengyan Shi, Giuseppe Castellucci, Simone Filice, Saar Kuzi, Elad Kravi, Eugene Agichtein, Oleg Rokhlenko, and Shervin Malmasi. 2025. Ambiguity detection and uncertainty calibration for question answering with large language models. In *Proceedings of the 5th Workshop on Trustworthy NLP (TrustNLP 2025)*. 41–55.
- [163] Ruihao Shui, Yixin Cao, Xiang Wang, and Tat-Seng Chua. 2023. A comprehensive evaluation of large language models on legal judgment prediction. *arXiv preprint arXiv:2310.11761* (2023).
- [164] Roger W Shuy. 2002. Linguistic battles in trademark disputes. (2002).
- [165] Marco Siino, Mariana Falco, Daniele Croce, and Paolo Rosso. 2025. Exploring LLMs Applications in Law: A Literature Review on Current Legal NLP Approaches. *IEEE Access* (2025).
- [166] Răzvan-Alexandru Smădu, Ion-Robert Dinică, Andrei-Marius Avram, Dumitru-Clementin Cercel, Florin Pop, and Mihaela-Claudia Cercel. 2022. Legal named entity recognition with multi-task domain adaptation. In *Proceedings of the Natural Legal Language Processing Workshop 2022*. 305–321.
- [167] Bogdana Stjepanovic. 2024. Leveraging artificial intelligence in ediscovery: enhancing efficiency, accuracy, and ethical considerations. *Regional L. Rev.* (2024), 179.
- [168] Benjamin Strickson and Beatriz De La Iglesia. 2020. Legal judgement prediction for UK courts. In *Proceedings of the 3rd International Conference on Information Science and Systems*. 204–209.
- [169] Weihang Su, Qingyao Ai, Yueyue Wu, Yixiao Ma, Haitao Li, and Yiqun Liu. 2023. Caseformer: Pre-training for legal case retrieval. *arXiv preprint arXiv:2311.00333* (2023).
- [170] Yu Sun, Shuohuan Wang, Shikun Feng, Siyu Ding, Chao Pang, Junyuan Shang, Jiaxiang Liu, Xuyi Chen, Yanbin Zhao, Yuxiang Lu, et al. 2021. Ernie 3.0: Large-scale knowledge enhanced pre-training for language understanding and generation. *arXiv preprint arXiv:2107.02137* (2021).
- [171] Zhongxiang Sun. 2023. A short survey of viewing large language models in legal aspect. *arXiv preprint arXiv:2303.09136* (2023).
- [172] Richard Susskind. 1996. The Future of Law, and Transforming the Law.
- [173] Yanran Tang, Ruihong Qiu, Yilun Liu, Xue Li, and Zi Huang. 2024. CaseGNN: Graph Neural Networks for Legal Case Retrieval with Text-Attributed Graphs. In *European Conference on Information Retrieval*. Springer, 80–95.
- [174] Nina Toren. 1975. Deprofessionalization and its sources: A preliminary analysis. *Sociology of work and occupations* 2, 4 (1975), 323–337.
- [175] Stephen E Toulmin. 2003. *The uses of argument*. Cambridge university press.
- [176] Vu Tran, Minh Le Nguyen, and Ken Satoh. 2019. Building legal case retrieval systems with lexical matching and summarization using a pre-trained phrase scoring model. In *Proceedings of the seventeenth international conference on artificial intelligence and law*. 275–282.
- [177] Santosh Tyss, Marcel Perez San Blas, Phillip Kemper, and Matthias Grabmair. 2023. Leveraging task dependency and contrastive learning for case outcome classification on european court of human rights cases. In *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics*. 1103–1111.
- [178] JAGM Van Dijk. 2017. Digital divide: Impact of access. *The international encyclopedia of media effects* 1 (2017), 1–11.
- [179] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. 2017. Attention is all you need. *Advances in neural information processing systems* 30 (2017).
- [180] Shaurya Vats, Atharva Zope, Somsubhra De, Anurag Sharma, Upal Bhattacharya, Shubham Kumar Nigam, Shouvik Guha, Koustav Rudra, and Kripabandhu Ghosh. 2023. Llms—the good, the bad or the indispensable?: A use case on legal statute prediction and legal judgment prediction on indian court cases. In *Findings of the Association for Computational Linguistics: EMNLP 2023*. 12451–12474.
- [181] Bart Verheij. 2017. Proof with and without probabilities: Correct evidential reasoning with presumptive arguments, coherent hypotheses and degrees of uncertainty. *Artificial Intelligence and Law* 25 (2017), 127–154.
- [182] Pepijn RS Visser and Trevor JM Bench-Capon. 1998. A comparison of four ontologies for the design of legal knowledge systems. *Artificial Intelligence and law* 6, 1 (1998), 27–57.
- [183] Charlotte S Vlek, Henry Prakken, Silja Renooij, and Bart Verheij. 2014. Building Bayesian networks for legal evidence with narratives: a case study evaluation. *Artificial intelligence and law* 22 (2014), 375–421.
- [184] Charlotte S Vlek, Henry Prakken, Silja Renooij, and Bart Verheij. 2016. A method for explaining Bayesian networks for legal evidence with scenarios. *Artificial Intelligence and Law* 24 (2016), 285–324.
- [185] Thi-Hai-Yen Vuong, Hai-Long Nguyen, Tan-Minh Nguyen, Hoang-Trung Nguyen, Thai-Binh Nguyen, and Ha-Thanh Nguyen. 2024. NOWJ at COLIEE 2023: Multi-task and Ensemble Approaches in Legal Information Processing. *The Review of Socionetwork Strategies* 18, 1 (2024), 145–165.
- [186] Douglas Walton. 2003. Is there a burden of questioning? *Artificial Intelligence and Law* 11, 1 (2003), 1–43.

- [187] Douglas Walton. 2010. Similarity, precedent and argument from analogy. *Artificial Intelligence and Law* 18 (2010), 217–246.
- [188] Douglas Walton. 2014. Baseballs and arguments from fairness. *Artificial intelligence and law* 22 (2014), 423–449.
- [189] Xuran Wang, Xinguang Zhang, Vanessa Hoo, Zhouhang Shao, and Xuguang Zhang. 2024. LegalReasoner: A Multi-Stage Framework for Legal Judgment Prediction via Large Language Models and Knowledge Integration. *IEEE Access* (2024).
- [190] Zheng Wang, Yuanzhi Ding, Caiyuan Wu, Yuzhen Guo, and Wei Zhou. 2024. Causality-inspired legal provision selection with large language model-based explanation. *Artificial Intelligence and Law* (2024), 1–25.
- [191] Sabine Wehnert, Shipra Dureja, Libin Kuttty, Viju Sudhi, and Ernesto William De Luca. 2022. Applying BERT embeddings to predict legal textual entailment. *The Review of Socionetwork Strategies* 16, 1 (2022), 197–219.
- [192] Bin Wei, Yaoyao Yu, Leilei Gan, and Fei Wu. 2025. An LLMs-based neuro-symbolic legal judgment prediction framework for civil cases. *Artificial Intelligence and Law* (2025), 1–35.
- [193] Laura Weidinger, John Mellor, Maribeth Rauh, Conor Griffin, Jonathan Uesato, Po-Sen Huang, Myra Cheng, Mia Glaese, Borja Balle, Atoosa Kasirzadeh, et al. 2021. Ethical and social risks of harm from language models. *arXiv preprint arXiv:2112.04359* (2021).
- [194] Johannes Welbl, Amelia Glaese, Jonathan Uesato, Sumanth Dathathri, John Mellor, Lisa Anne Hendricks, Kirsty Anderson, Pushmeet Kohli, Ben Coppin, and Po-Sen Huang. 2021. Challenges in detoxifying language models. *arXiv preprint arXiv:2109.07445* (2021).
- [195] Nirmalie Wiratunga, Ramitha Abeyratne, Lasal Jayawardena, Kyle Martin, Stewart Massie, Ikechukwu Nkisi-Orji, Ruwan Weerasinghe, Anne Liret, and Bruno Fleisch. 2024. CBR-RAG: case-based reasoning for retrieval augmented generation in LLMs for legal question answering. In *International Conference on Case-Based Reasoning*. Springer, 445–460.
- [196] Jerzy Wróblewski. 1974. Legal syllogism and rationality of judicial decision. *Rechtstheorie* 5 (1974), 33.
- [197] Xingyu Wu, Sheng-hao Wu, Jibin Wu, Liang Feng, and Kay Chen Tan. 2025. Evolutionary computation in the era of large language model: Survey and roadmap. *IEEE Transactions on Evolutionary Computation* 29, 2 (2025), 534–554.
- [198] Xingyu Wu, Kui Yu, Jibin Wu, and Kay Chen Tan. 2025. LLM Cannot Discover Causality, and Should Be Restricted to Non-Decisional Support in Causal Discovery. *arXiv preprint arXiv:2506.00844* (2025).
- [199] Yangbin Xia and Xudong Luo. 2024. Legal Judgment Prediction with LLM and Graph Contrastive Learning Networks. In *Proceedings of the 2024 8th International Conference on Computer Science and Artificial Intelligence*. 424–432.
- [200] Chaojun Xiao, Xueyu Hu, Zhiyuan Liu, Cunchao Tu, and Maosong Sun. 2021. Lawformer: A pre-trained language model for chinese legal long documents. *AI Open* 2 (2021), 79–84.
- [201] Zhuopeng Xu, Xia Li, Yinlin Li, Zihan Wang, Yujie Fanxu, and Xiaoyan Lai. 2020. Multi-task legal judgement prediction combining a subtask of the seriousness of charges. In *Chinese Computational Linguistics: 19th China National Conference, CCL 2020, Hainan, China, October 30–November 1, 2020, Proceedings* 19. Springer, 415–429.
- [202] Shuxin Yang, Suxin Tong, Guixiang Zhu, Jie Cao, Youquan Wang, Zhengfa Xue, Hongliang Sun, and Yu Wen. 2022. MVE-FLK: A multi-task legal judgment prediction via multi-view encoder fusing legal keywords. *Knowledge-Based Systems* 239 (2022), 107960.
- [203] Xiaoxian Yang, Zhifeng Wang, Qi Wang, Ke Wei, Kaiqi Zhang, and Jiangang Shi. 2024. Large language models for automated q&a involving legal documents: a survey on algorithms, frameworks and applications. *International Journal of Web Information Systems* 20, 4 (2024), 413–435.
- [204] Masaharu Yoshioka, Youta Suzuki, and Yasuhiro Aoki. 2022. Hukb at the coliee 2022 statute law task. In *JSAT International Symposium on Artificial Intelligence*. Springer, 109–124.
- [205] Shengbin Yue, Wei Chen, Siyuan Wang, Bingxuan Li, Chenchen Shen, Shujun Liu, Yuxuan Zhou, Yao Xiao, Song Yun, Xuanjing Huang, et al. 2023. Disc-lawllm: Fine-tuning large language models for intelligent legal services. *arXiv preprint arXiv:2309.11325* (2023).
- [206] Tomasz Zalewski. 2021. Basic Principles for the Effective Use of Legal Tech Tools. In *Legal Tech*. Nomos Verlagsgesellschaft mbH & Co. KG, 315–332.
- [207] Rowan Zellers, Ari Holtzman, Hannah Rashkin, Yonatan Bisk, Ali Farhadi, Franziska Roesner, and Yejin Choi. 2019. Defending against neural fake news. *Advances in neural information processing systems* 32 (2019).
- [208] Guohang Zeng, George Tian, Guangquan Zhang, and Jie Lu. 2025. RoSiLC-RS: A Robust Similar Legal Case Recommender System empowered by large language model and step-back prompting. *Neurocomputing* (2025), 130660.
- [209] Jian Zeng, Kaixin Chen, Ruiqi Wang, Yilong Li, Mingming Fan, Kaishun Wu, Xiaoke Qi, and Lu Wang. 2025. Contract-Mind: Trust-calibration interaction design for AI contract review tools. *International Journal of Human-Computer Studies* 196 (2025), 103411.
- [210] Han Zhang, Zhicheng Dou, Yutao Zhu, and Ji-Rong Wen. 2023. Contrastive learning for legal judgment prediction. *ACM Transactions on Information Systems* 41, 4 (2023), 1–25.

- [211] Yue Zhang, Zhiliang Tian, Shicheng Zhou, Haiyang Wang, Wenqing Hou, Yuying Liu, Xuechen Zhao, Minlie Huang, Ye Wang, and Bin Zhou. 2025. RLJP: Legal Judgment Prediction via First-Order Logic Rule-enhanced with Large Language Models. *arXiv preprint arXiv:2505.21281* (2025).
- [212] Zhengyan Zhang, Xu Han, Zhiyuan Liu, Xin Jiang, Maosong Sun, and Qun Liu. 2019. ERNIE: Enhanced language representation with informative entities. *arXiv preprint arXiv:1905.07129* (2019).
- [213] Lucia Zheng, Neel Guha, Brandon R Anderson, Peter Henderson, and Daniel E Ho. 2021. When does pretraining help? assessing self-supervised learning for law and the casehold dataset of 53,000+ legal holdings. In *Proceedings of the eighteenth international conference on artificial intelligence and law*. 159–168.
- [214] Haoxi Zhong, Chaojun Xiao, Cunchao Tu, Tianyang Zhang, Zhiyuan Liu, and Maosong Sun. 2020. How does NLP benefit legal system: A summary of legal artificial intelligence. *arXiv preprint arXiv:2004.12158* (2020).
- [215] Zhi Zhou, Jiang-Xin Shi, Peng-Xiao Song, Xiao-Wen Yang, Yi-Xuan Jin, Lan-Zhe Guo, and Yu-Feng Li. 2024. Lawgpt: A chinese legal knowledge-enhanced large language model. *arXiv preprint arXiv:2406.04614* (2024).

Received 20 February 2025; revised 12 March 2025; accepted 5 June 2025